

Federated Domain Generalization: A Survey

This article serves as the first survey on the topic of FDG, aiming to introduce recent advances with an emphasis on its theories, methodologies, datasets, applications, evaluations, benchmarks, and future research directions.

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ABSTRACT | Machine learning (ML) typically relies on the assumption that training and testing distributions are identical and that data are centrally stored for training and testing. However, in real-world scenarios, distributions may differ significantly, and data are often distributed across different devices, organizations, or edge nodes. Consequently, it is to develop models capable of effectively generalizing across unseen distributions in data spanning various domains. In response to this challenge, there has been a surge of interest in federated domain generalization (FDG) in recent years. FDG synergizes federated learning (FL) and domain generalization (DG) techniques, facilitating collaborative model development across diverse source domains for effective generalization to

unseen domains, all while maintaining data privacy. However, generalizing the federated model under domain shifts remains a complex, underexplored issue. This article provides a comprehensive survey of the latest advancements in this field. Initially, we discuss the development process from traditional ML to domain adaptation (DA) and DG, leading to FDG, as well as provide the corresponding formal definition. Subsequently, we classify recent methodologies into four distinct categories: federated domain alignment (FDAL), data manipulation (DM), learning strategies (LSs), and aggregation optimization (AO), detailing appropriate algorithms for each. We then overview commonly utilized datasets, applications, evaluations, and benchmarks. Conclusively, this survey outlines potential future research directions.

KEYWORDS | Domain generalization (DG); domain shift; federated domain generalization (FDG); machine learning (ML); privacy preserving.

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NOMENCLATURE

ML	Machine learning.
FL	Federated learning.
DA	Domain adaptation.
FDA	Federated domain adaptation.
DG	Domain generalization.
FDG	Federated domain generalization.
DIR	Domain-invariant representation.
MMD	Maximum mean discrepancy.
DANN	Domain adversarial neural network.
UDA	Unsupervised domain adaptation.
WD	Wasserstein distance.
NLP	Natural language processing.
MTSSL	Multitask self-supervised learning.
IID	Independent and identically distributed.
IoT	Internet of Things.
OOD	Out of distribution.

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FID	Fréchet inception distance.
LFRL	Lifelong federated reinforcement learning.
GAN	Generative adversarial network.
IRM	Invariant risk minimization.
KL	Kullback–Leibler divergence.
KD	Knowledge distillation.
MAML	Model-agnostic meta-learning.
mIoU	Mean intersection over union.
$\mathcal{N}$	Set of clients.
N	Number of clients in FL.
$\mathcal{M}$	Set of end users in FL.
M	Number of end users in FL.
$\mathcal{S}_i$	Dataset of client i .
$ \mathcal{S}_i $	Size of local dataset of client i .
$\mathcal{H}$	Hypothesis space.
h	Labeling function from a $\mathcal{H}$.
$\mathcal{X}$	Nonempty input space.
$\mathcal{Y}$	Output space.
P_{XY}	Probability distribution of the joint input–output space.
S	Domain.
M	Number of Domains.
$\mathcal{S}_{\text{source}}$	Set of the source domains.
$\mathcal{S}_{\text{target}}$	Set of the target domains.
M	Number of source domains.
n	Size of the domain.
$\mathcal{S}^i$	Domain i .
n_i	Size of the domain i .
$\mathcal{X}_i$	Selected subset from the local dataset $\mathcal{S}_i$ of client i for local model training.
$ \mathcal{X}_i $	Size of the chosen subset $\mathcal{X}_i$ from client i .
ω_t	Aggregated global model at round t .
ω_{t+1}^i	Local model from client i at round $t + 1$.
ω^*	Optimal weights of the global model.
η	Learning rate of FL training.
$\theta(h, \hat{h})$	Difference in the source domain between two labeling function h and $\hat{h}$.
$\theta^s(h)$	Risk of labeling function h on the source domains.
$\theta^t(h)$	Risk of labeling function h on the target domains.
$\lambda_{\mathcal{H}}$	Complexity of $\mathcal{H}$ for prediction tasks.
d	Vapnik–Chervonenkis dimension of $\mathcal{H}$.
$\mathbb{E}$	Expectation operator.
$\mathcal{U}^s$	Unlabeled samples of size k from source domain.
$\mathcal{U}^t$	Unlabeled samples of size k from target domain.
$\hat{\psi}(h)$	Estimation of the average risk over all possible target domains.
π_i	π Normalized mixing weights.
$F^t((x_i^t))$	Feature representation of the target domain data.
$\text{MMD}^2(P, Q)$	Square of maximum mean discrepancy.

I. INTRODUCTION

Humans possess an extraordinary aptitude to transfer their knowledge and expertise in novel situations and environments they have not experienced before, while machines struggle to reproduce this proficiency, especially regarding OOD data [1]. This raises pertinent questions regarding the efficacy of ML models across different contexts. For instance, can a model train to detect tumors in medical images obtained from one type of scanner maintain its effectiveness when applied to images from another scanner or imaging protocol? Similarly, does a robot trained to execute a task within a simulated environment possess the capability to generalize its skills to the real world? Additionally, there is a concern about whether a speech recognition system trained on audio recordings from one speaker would perform well when tested on recordings from a different speaker or whether a language model trained on text from one domain (e.g., news articles) would apply to text from a different domain (e.g., social media). The resolution of these questions hinges on the ML models' ability to navigate the prevalent issue of domain/dataset shift [2]. Domain shift refers to the discrepancy in distribution between the source domain employed for training and the target domain utilized for testing [3]. It is a pervasive issue in ML, significantly impacting the models' generalization and robustness across unseen domains in real-world scenarios.

Recently, deep learning (DL) has made revolutionary advances and emerged as a powerful tool in a variety of fields, including image recognition, NLP, speech recognition, and robotics, among others. However, the prevalent assumption in DL algorithms is that all samples in both training and test datasets are IID [4]—an assumption that often does not hold in OOD scenarios encountered in real-world applications. This discrepancy, known as domain shift, can significantly degrade a model's performance when applied to unseen domains. To tackle the issue of domain shift, the technique of DG [1], [5] has been developed, enhancing model generalization across disparate domains, irrespective of their similarity to the training domain.

There has been significant research in the area of DG, and numerous strategies have been developed to tackle the issue of domain shift. Predominantly, DG methods have been centralized, utilizing a central server that accesses data from all source domains for DG tasks. Specifically, domain alignment methods [6], [7] aim to align feature distributions between the source domains and target domains, thereby improving model generalization. This approach directly tackles the issue of domain shift by ensuring that the model perceives the source and target domains as similar, thereby facilitating improved performance on unseen domains. Data augmentation techniques [8], [9] seek to expand the training dataset's size and diversity, enhancing adaptability without new

data collection. Data augmentation acts as a complementary strategy to domain alignment by enriching the model's training environment, thereby indirectly reducing the impact of domain discrepancies. Meta-learning strategies [10], [11] focus on optimizing the model's initialization or optimization process for quick adaptation to new tasks or domains with minimal additional training. This approach addresses the generalization issue by preparing the model to efficiently learn from limited data in new domains, rather than directly aligning domain features or augmenting data.

Ensemble learning methods [12], [13] synthesize predictions from models trained on various domains, using ensemble techniques to identify common patterns and enhance generalization to unseen domains. This technique enhances model robustness and generalization by aggregating insights from multiple perspectives, thereby mitigating the risk of overfitting to domain-specific features and elevating performance on unseen domains. However, these methods may significantly infringe upon data protection laws like the EU/U.K. General Data Protection Regulation [14], as tremendous privacy-containing data are stored in locally distributed locations due to the rise of ML and the proliferation of IoT devices [15]. For instance, data collected from public transportation systems, ride-sharing platforms, and logistics companies contribute to the transportation sector; data from banks, insurance providers, and credit rating agencies pertain to the financial industry; sensitive patient information is managed by healthcare institutions; and educational records are maintained by schools, universities, and online learning platforms. Therefore, developing a highly generalizable model in real-world scenarios that can ensure both privacy and high model performance is a dilemma [16]. On the one hand, it necessitates careful consideration of privacy constraints across heterogeneous domains. On the other hand, the lack of simultaneous access to data from source domains hinders the accurate identification and extraction of domain-invariant features, which are essential for improving generalization.

Resolving the above dilemma necessitates a thorough comprehension of FL [17]. As a promising paradigm within distributed ML, FL facilitates the training of algorithms across a wide array of decentralized edge clients. Each of these clients retains its local data samples, thereby preventing the need for direct data exchange. This approach contrasts with traditional centralized learning methods, which necessitates aggregating data on a single server. FL primarily aims to utilize rich and diverse data sources, ensuring user privacy and reducing the need for extensive data transmission. Building upon this, FDG [18], [19] leverages FL alongside DG techniques to enable multiple source domains to collaboratively learn a model that adeptly generalizes well to unseen domains while keeping their data private. However, the technical complexity of achieving model generalization across domain shifts in FL settings remains a challenge with limited research focus

thus far. Given the evolving landscape of DG research, the introduction of the FL, characterized by distributed data domains, presents new challenges that necessitate innovative approaches. As stated above, most conventional solutions achieve DG by accessing multisource domains in a centralized manner, while each client can only access its local data in FDG. In addition, local optimization of the FL model may result in a bias toward its data distribution, which reduces its generalizability to new target domains. As a result, conventional DG methods are unsuitable for the unique requirements of FDG scenarios.

This article presents the first comprehensive survey on the topic of FDG, aiming to introduce recent advances in this field with an emphasis on its theories, methodologies, datasets, applications, evaluations, benchmarks, and future research directions. To our knowledge, there have been few recent attempts [1], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30] to explore this area, but their focus differs significantly from the scope of our survey. Table 1 provides a summary of comparisons between existing but related surveys and our work. Specifically, Bonawitz et al. [20] introduced a system design for FL on mobile devices utilizing TensorFlow, addressing the challenges and open questions in mobile phone domains. Yang et al. [21] developed a secure FL framework as a solution to the obstacles faced by AI in industries where data exist in isolated islands, and data privacy and security are strengthened. The framework includes horizontal FL, vertical FL, and federated transfer learning (FTL) and can be applied to various businesses successfully. Li et al. [22] explored the principles of FL, which entails training statistical models on remote devices or in siloed data centers while maintaining data localization. They elaborated on the intrinsic challenges of FL, summarizing existing strategies and articulating avenues for future investigation. Kairouz et al. [23] surveyed recent advancements, pinpointing the challenges and open problems in FL research.

Kouw and Loog [25] conducted a comprehensive review of recent advancements in DA, examining the fundamental question: how can a classifier effectively learn from a source domain and generalize to a target domain? They presented different approaches to solving the problem of DA in ML and they are categorized into sample-, feature-, and inference-based methods. Csurka [24] provided an overview of DA and transfer learning in visual applications. They discuss the state-of-the-art methods for different scenarios, including shallow and deep methods, and relate DA to other ML solutions. Wilson and Cook [26] conducted a comprehensive survey of methods used in single-source unsupervised deep DA, which integrates DL with DA techniques to reduce dependence on costly target domain annotations. Their work provides a detailed comparative analysis of various approaches, focusing on evaluating alternative methods, identifying unique and shared components, examining empirical results and theoretical foundations, and exploring potential applications and future research directions.

Table 1 Summary and Comparison of Key Surveys Across Related Research Paradigms, Detailing Their Publication Year, Domain Focus, and Thematic Scope

Ref.	Year	Domain	Scope
Bonawitz et al. [20]	2019	FL	A scalable production system for FL in the domain of mobile devices.
Yang et al. [21]	2019	FL	A comprehensive secure federated learning framework that includes horizontal, vertical, and transfer learning.
Li et al. [22]	2020	FL	Unique characteristics, challenges, current approaches, and future research areas.
Kairouz et al. [23]	2021	FL	Recent advancements, the challenges and open problems.
Csurka et al. [24]	2017	DA	An overview of domain adaptation and transfer learning in visual applications
Kuw et al. [25]	2019	DA	Sample-based, feature-based, and inference-based methods.
Wilson et al. [26]	2020	DA	A unsupervised domain adaptation methods in deep learning.
Wang et al. [27]	2018	DG	Data manipulation, representation learning, and learning strategy.
Zhou et al. [1]	2022	DG	Background, existing methods and theories, and insights and discussions on future research directions
Sheth et al. [28]	2022	DG	Invariance via Causal Data Augmentation, Invariance via Causal representation learning, and Invariance via Transferring Causal mechanisms.
Shen et al. [29]	2021	OOD Generalization	Definition, methodology, evaluation, and implications of Out-of-Distribution generalization.
Li et al. [30]	2022	OOD Generalization	Existing methods, theories, and future research directions related to out-of-Distribution generalization.
Our Survey	2025	FDG	Theories, methodologies, datasets, applications, evaluations, benchmarks, and future research directions.

DG extends the ideas behind DA by preparing a model to generalize across unseen target domains without requiring access to their data during the training phase. Wang et al. [27] reviewed recent advances in DG, which deals with the call into question of developing models capable of generalizing to unseen test domains. They also discussed the theories, algorithms, datasets, applications, and potential research topics related to DG. Zhou et al. [1] provided a comprehensive literature review of DG in ML, in which the review covers the background, existing methods and theories, and potential future research directions. Sheth et al. [28] discussed the problem of distribution shifts in ML models and how it affects their generalization capabilities. They survey various DG methods that leverage causality to identify stable features that remain invariant across different distributions. OOD generalization broadens the concept of DG to address the challenge of ensuring model robustness when encountering data that significantly deviate from the training distribution, including entirely new scenarios that were not represented in the training data. Liu et al. [29] discussed the OOD generalization problem in ML and provided a comprehensive review of existing methods, evaluation metrics, and future research directions in OOD generalization. The primary objective is to develop and rigorously assess models and approaches capable of adapting to distributional changes, demonstrating robust generalization, and resolving the prevalent issue of inconsistencies in data distributions encountered in real-world settings. Li et al. [30] reviewed recent advances in OOD generalization on graphs, which goes beyond the in-distribution hypothesis and presents a detailed categorization of methodologies into three categories based on their integration points within the graph ML pipeline.

FDG can be seen as an extension of OOD generalization to the FL paradigm, where the call into question is not only to handle unseen distributions but also to do so in a manner that is compatible with the decentralized and privacy-preserving principle of FL. Despite the notable advancements in FL and DG across various fields recently,

a comprehensive survey that cohesively covers theories, methodologies, datasets, applications, evaluations, benchmarks, and future research directions in FDG has yet to be undertaken. Our objective is to furnish researchers with an in-depth review of the current landscape and to catalyze further investigation in FDG and related fields. To achieve this, we have conducted an extensive review of the existing literature and synthesized the main ideas and contributions of the most relevant works. We aim for this survey to serve as a valuable resource for researchers aiming to expand their knowledge of FDG and to foster progress in this critical area.

The organization of this article is outlined as follows and illustrated in Fig. 1. The frequently used acronyms are summarized in Nomenclature. We begin by providing preliminaries to the field of FL, traditional ML, DA, DG, and FDG in Section II. Then, we introduce existing methodologies for FDG, including federated domain alignment (FDAL), data manipulation (DM), learning strategies (LSs), and aggregation optimization (AO) algorithms in Section III. Next, we present the applications in Section IV. To enhance clarity and conciseness, the detailed descriptions of the datasets, evaluations, and benchmarks discussed throughout the survey are provided in Appendixes A–C, respectively. Section V summarizes the insights derived from existing research and engages in a discussion concerning the future research directions in the field. Finally, a comprehensive conclusion of this article is presented in Section VI, summarizing the key findings and insights derived throughout the study. On the analysis of the sections, we derive a generalized architecture and workflow of FDG, as depicted in Fig. 2.

II. PRELIMINARIES

In this section, we begin by providing an overview of FL, DA, DG, and FDG. Subsequently, we discuss the integration of FL and DG. For the convenience of readers, we have summarized frequently used notations in Nomenclature.

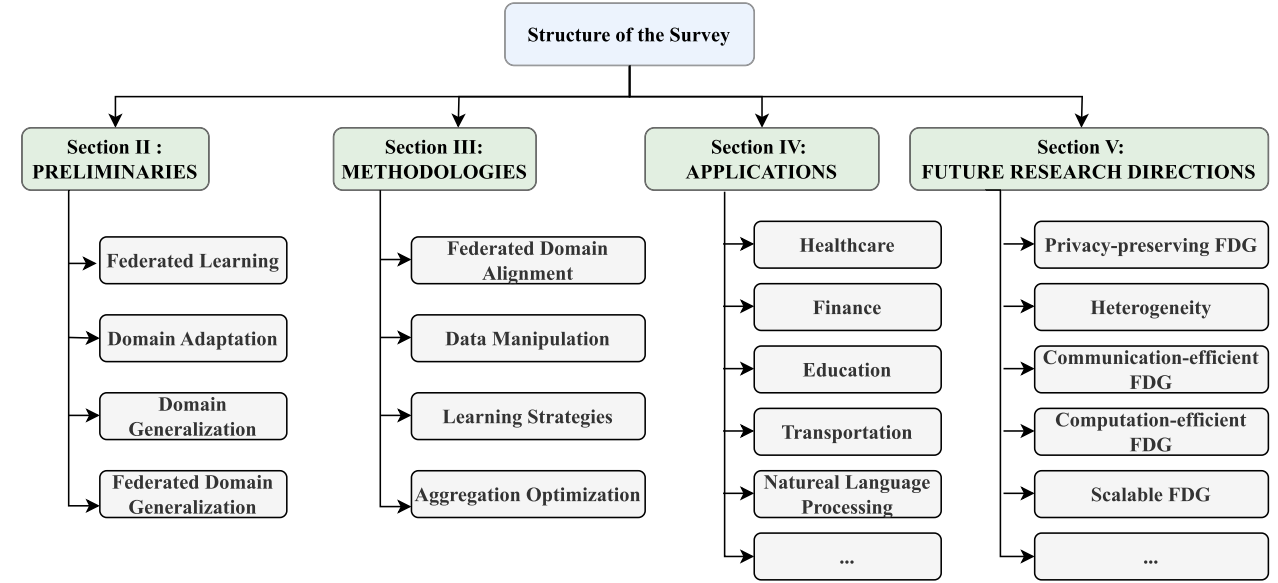


Fig. 1. Content structure of this article. The figure outlines the organization of this article and reflects the comprehensive scope of FDG.

A. Federated Learning

Artificial intelligence has swiftly emerged as a fundamental element of our daily lives, fueled by notable

advancements and breakthroughs in DL techniques across various domains. Moreover, the widespread adoption of IoT devices has resulted in enormous amounts of data

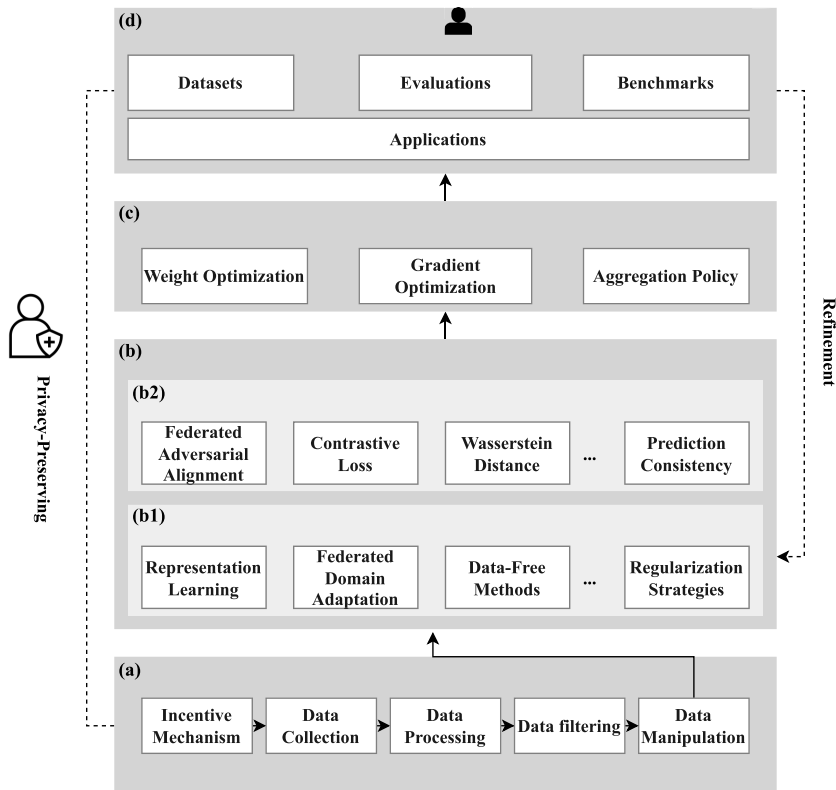


Fig. 2. Generalized architecture and workflow of FDG. The workflow consists of four stages. (a) Data preparation, involving the organization of decentralized data. (b) Local model training, including (b1) LSs for enhancing generalization and (b2) domain alignment techniques to mitigate interclient distribution shifts. (c) Aggregation, where local models are integrated into a global model while preserving domain-invariant features. (d) Model evaluation, which measures generalization and robustness on unseen domains.

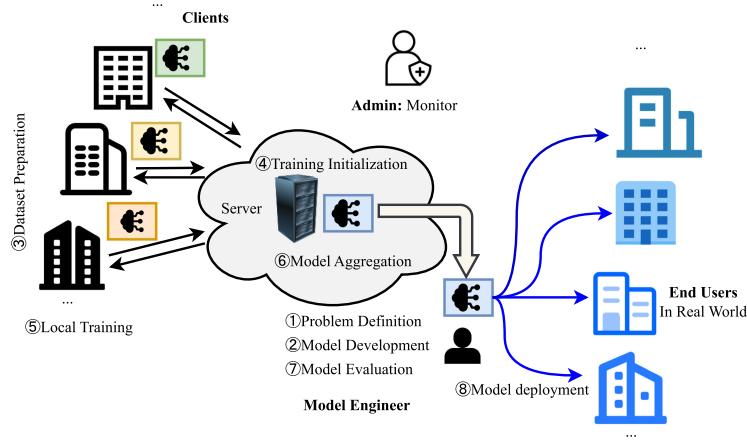


Fig. 3. Lifecycle of a trained model in FL. The process includes: ① problem definition; ② model development; ③ dataset preparation; ④ training initialization; ⑤ local training; ⑥ model aggregation; ⑦ model evaluation; and ⑧ model deployment to end users. This workflow outlines the end-to-end pipeline from model design to real-world application in FL systems.

being stored in locally distributed locations. In addition, with the proliferation of sensitive and private data, concerns over data privacy have become increasingly prominent. In response to these concerns, FL has been introduced as a solution to uphold data privacy concerns while facilitating collaborative ML.

As illustrated in Fig. 3, we depict a scenario consisting of a model engineer, an admin, a central server, N clients, and M end users. The set of clients is denoted by $\mathcal{N} = \{1, 2, \dots, i, \dots, N\}$, while the set of end users is denoted by $\mathcal{M} = \{1, 2, \dots, j, \dots, M\}$. Each client $i \in \mathcal{N}$ possesses a local dataset $\mathcal{S}^i$, which may change over time. The dataset's size, denoted by $|\mathcal{S}^i|$, allows clients to engage iteratively in FL processes, reflecting the dynamic nature of their datasets. For local model training, a client may select a subset $\mathcal{X}^i \subseteq \mathcal{S}^i$, with its size represented by $|\mathcal{X}^i|$. Next, we will introduce the lifecycle, key actors, and typical processes within an FL system.

1) Lifecycle of a Trained Model in FL System: The FL process is typically driven by a model engineer who develops a model for a particular application. Here is a high-level overview of a typical FL workflow.

Step 1 (Problem Definition): The first step in the FL workflow is to define the problem that the model will solve. This involves identifying the data sources and the target task, as well as specifying the performance metrics that will be used to evaluate the model.

Step 2 (Model Development): Once the problem is defined, the model engineer develops a model architecture that is suitable for the FL paradigm. This involves selecting the appropriate optimization algorithm, defining the model parameters, and deciding how to partition the data among the participating clients.

Step 3 (Dataset Preparation): The dataset is prepared by the data owners, who collect or generate the data that will be used for training the model. The

data are partitioned among the clients according to the protocol defined in Step 2.

Step 4 (Training Initialization): The central server initializes the training process by sending the initial model weights to the participating clients. This model can be randomly initialized or pretrained on a large dataset.

Step 5 (Local Training): The participating clients perform local training using their own data and the initial model weights. This training is typically done using stochastic gradient descent or a variant of it.

Step 6 (Model Aggregation): After local training, the clients return their updated model weights to the central server, which aggregates them using an algorithm such as FedAvg or other optimized aggregation algorithms.

Step 7 (Model Evaluation): The engineer evaluates the aggregated model on a validation dataset to determine its accuracy and generalization performance. If the performance is unsatisfactory, the training process can be repeated with a different set of initial weights or hyperparameters.

Step 8 (Model Deployment): Once the model is trained, it can be deployed to predict unseen domains. In some cases, the models need to be fine-tuned on new data to adapt to changing conditions.

Overall, the workflow involves multiple iterations of local training, model aggregation, and evaluation, and it requires close collaboration between the model engineer and the participating clients. The success of an FL project depends on careful planning, effective communication, and the ability to adapt to changing conditions and requirements.

2) Key Actors in FL System: FL encompasses a variety of stakeholders, each contributing distinct roles in the model's training and deployment phases. The following outlines the key actors within an FL framework.

- 1) *Model Engineer*: The model engineer plays a critical role in the FL, tasked with the development of the ML model and the establishment of the training protocol. The role involves selecting an appropriate model architecture, optimization algorithm, and partitioning strategy for the distributed data. The model engineer also evaluates the model's performance and performs fine-tuning as needed.
- 2) *Admin*: In FL, the admin is a role that involves managing the overall operation of the FL system. An admin may be an individual or a team responsible for configuring and monitoring the system, as well as ensuring its security, privacy, and compliance with legal and ethical requirements.
- 3) *Central Server*: The central server coordinates the FL training process by collecting and aggregating the model parameters from the participating clients. The server performs the model aggregation using algorithms.
- 4) *Participating Clients*: They are the devices or organizations that contribute their local data and computational resources to the FL training process. They perform local training on their data and share their model parameters with the central server for aggregation.
- 5) *End Users*: They, comprising individuals or organizations, utilize the trained model for predictions or decision-making in novel or previously unseen domains. They may include consumers, businesses, or government agencies that benefit from the insights generated by the model. Depending on the application requirements, the trained model may be deployed on mobile devices, cloud servers, or edge devices.
- 6) *Communication Channel*: It is the medium used for transmitting the model weights between the central server and the participating clients. It can be a wireless or wired network connection and must be designed to ensure the privacy and security of the data transmitted between the parties involved.

Overall, an FL system involves multiple actors collaborating to train and deploy an ML model. The success of an FL project depends on the effective coordination and communication between these actors, as well as the careful consideration of privacy, security, and performance concerns.

3) *Typical Process of FL Training*: FL endeavors to construct a robust global model by harnessing insights from distributed data without compromising privacy. The objective of FL is to achieve a global model ω^* that optimally minimizes the global loss function $L(\omega)$ across all participating clients. Achieved through iterative local training and aggregation, this process enables learning from varied data distributions without direct data access. The typical FL training process unfolds through several essential steps:

Step 1 (Client Selection): In FL, client selection is a pivotal step that determines participant involvement in the training process. The objective is to identify a subset of clients, denoted as $\mathcal{N}^s$ from all clients $\mathcal{N}$, where $\mathcal{N}^s \subseteq \mathcal{N}$, ensuring a representative set that enriches the global model without compromising data privacy or overextending computational resources. Selection strategies may vary from random to more sophisticated methods tailored to identify suitable participants. This is imperative as inappropriate clients, such as poor data quality, limited diversity in data, heightened privacy risks, substantial resource constraints, or data that poorly align with project goals, can detract from the process's effectiveness, security, and efficiency. Consequently, meticulous client selection is essential to preserve the integrity of the global model and secure successful training outcomes.

Step 2 (Download): In this step, the clients first need to download the global model aggregated by the central server in the previous round t (the global model is randomly initialized in the first round)

$$\omega_t^s = \omega_t \quad (1)$$

where ω_t denotes the global model in the round t .

Step 3 (Local Training): The participating clients perform local training on their respective domains using the downloaded model

$$\omega_{t+1}^i = \omega_t^i - \eta \nabla L^i(\omega_t^i; \mathcal{X}^i) \quad (2)$$

where η denotes the learning rate, $L^i(\omega_t^i; \mathcal{X}^i)$ represents the local loss function for client i at round t , and ω_{t+1}^i represents the updated local model parameters for client i at round $t + 1$.

Step 4 (Upload): The locally trained models are then transmitted to the central server for aggregation.

Step 5 (Model Aggregation): The central server aggregates the model parameters from the participating clients using a suitable algorithm. The aggregation process aims to combine the local model updates into a single global model that reflects the collective knowledge of all clients

$$\omega_{t+1} = \frac{\sum_{i \in \mathcal{N}^s} |\mathcal{X}^i| \omega_{t+1}^i}{\sum_{i \in \mathcal{N}^s} |\mathcal{X}^i|} \quad (3)$$

where ω_{t+1} denotes the aggregated global model in the round $t + 1$.

Step 6 (Iteration): Return to Step 2 until the convergence criteria are met [31], and then, terminate the training. The iterative nature of this process ensures continuous refinement of the model, leading to the optimal global model ω^* that minimizes the global

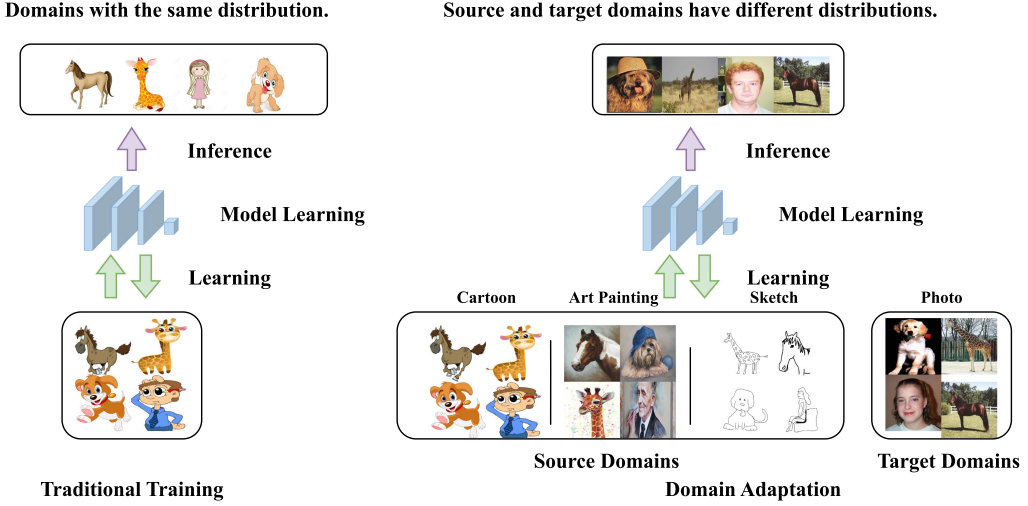


Fig. 4. Comparison of traditional ML tasks (left) and DA tasks (right). In traditional ML, training and test data are drawn from the same distribution. In DA, the model is trained on source domains and applied to target domains with different distributions, requiring techniques to bridge the domain gap.

loss function $L(\omega)$

$$L(\omega) = \frac{\sum_{i \in \mathcal{N}^s} |\mathcal{X}^i| L^i(\omega; \mathcal{X}_i)}{\sum_{i \in \mathcal{N}^s} |\mathcal{X}^i|} \quad (4)$$

where $L^i(\omega; \mathcal{X}_i)$ is the loss for subset $\mathcal{X}_i$ given ω

$$\omega^* = \arg \min_{\omega} L(\omega). \quad (5)$$

This process not only safeguards data privacy but also leverages diverse data distributions to enhance the model's accuracy and generalizability. The culmination of this rigorously designed process is a global model that encapsulates the collective knowledge of all clients, finely tuned to achieve the common goal of minimizing the global loss function.

B. Domain Adaptation

Recent studies [32], [33] have indicated that DL models are vulnerable to significant performance degradation when evaluated on OOD datasets, even with minor variations in the data generation process. Specifically, it demonstrated that DL models exhibit limited generalization, which can significantly impact their performance when deployed in real-world scenarios. DA [34], [35] emerged as a strategy to circumvent the OOD data challenge by collecting a subset of data from target domains, thereby enabling the adaptation of a model trained on a source domain with disparate distributions to the target domain. Fig. 4 compares traditional ML tasks and DA tasks. The basic idea behind DA is that, despite variations between source and target domains, shared underlying patterns can be exploited to enhance model performance in the target domain, as delineated in Definition 2.

Definition 1 (Domain): A domain refers to a collection of data that are sampled from a joint distribution. Consider a nonempty input space denoted as $\mathcal{X}$ and an output space denoted as $\mathcal{Y}$. The domain, denoted as $\mathcal{S}$, comprises n samples $\mathcal{S} = \{(x_j, y_j)_{j=1}^n \sim P_{XY}\}$, where $x_j \in \mathcal{X} \subset \mathbb{R}^d$ represents a d -dimensional input sample, $y_j \in \mathcal{Y} \subset \mathbb{R}$ represents the corresponding output label, and P_{XY} denotes the probability distribution over the joint input-output space, where X and Y represent the respective random variables.

Definition 2 (DA): Let h represent any labeling function from the hypothesis space $\mathcal{H}$. Given M source domains denoted as $\mathcal{S}_{\text{source}} = \{\mathcal{S}^i | i = 1, 2, \dots, M\}$, where $\mathcal{S}^i = \{(x_j^i, y_j^i)_{j=1}^{n_i}\}$ represents the i th domain with true labeling functions h^{*s} , it is observed that the joint distributions between each pair of domains differ ($P_{XY}^i \neq P_{XY}^{\hat{i}}$, for $1 \leq i \neq \hat{i} \leq M$). The goal of DA is to reduce the disparity between the source domains and target domains, enabling effective generalization on the target domain $\mathcal{S}_{\text{target}}$, where $\mathcal{S}_{\text{target}}$ is accessible during training but has a distinct distribution ($P_{XY}^{\text{target}} \neq P_{XY}^i$ for $i \in 1, 2, \dots, M$), with corresponding true labeling functions denoted as h^{*t} .

Therefore, the difference between two labeling functions, h and $\hat{h}$, within the source domain can be quantified as follows:

$$\begin{aligned} \theta^s(h, \hat{h}) &= \mathbb{E}_{x \sim P_X^s} [h(x) \neq \hat{h}(x)] \\ &= \mathbb{E}_{x \sim P_X^s} [|h(x) - \hat{h}(x)|]. \end{aligned} \quad (6)$$

Similarly, we can get θ^t when $x \sim P_X^t$. Let $\theta^s(h) := \theta^s(h, h^{*s})$ and $\theta^t(h) := \theta^t(h, h^{*t})$ denote the risk associated with labeling function h in the source domain and target domain, respectively.

DA seeks to reduce the target domain risk, $\theta^t(h)$. However, directly assessing this risk is often impractical due to the lack of explicit knowledge about the target domain's true distribution. To overcome the issue, Ben-David et al. [36] proposed a theoretical framework to approximate the target domain risk, $\theta^t(h)$, using the more accessible source domain risk, $\theta^s(h)$

$$\theta^t(h) \leq \theta^s(h) + d_{\mathcal{H}\Delta\mathcal{H}}(P_X^s, P_X^t) + \lambda_{\mathcal{H}} \quad (7)$$

where the discrepancy between cross-domain distributions is quantified by the $\mathcal{H}\Delta\mathcal{H}$ -divergence $d_{\mathcal{H}\Delta\mathcal{H}}(P_X^s, P_X^t)$, while the complexity of $\mathcal{H}$ for prediction tasks on the source domain and target domain is assessed by the ideal joint risk $\lambda_{\mathcal{H}}$. Notably, the $\mathcal{H}\Delta\mathcal{H}$ -divergence offers superior finite sample guarantees compared to other divergence measures and allows for nonasymptotic bounds. The nonasymptotic bound ensures that the bound holds a high probability, regardless of the sample size, which is particularly useful in applications where the sample size is small or limited. This means that with a finite sample size, the $\mathcal{H}\Delta\mathcal{H}$ -divergence can provide a more accurate estimate of the difference between two probability distributions than other measures.

Theorem 1 (DA Error Bound (Nonasymptotic) [27], [36]): Let $\mathcal{U}^s$ and $\mathcal{U}^t$ represent unlabeled samples of size k from the two domains. The Vapnik–Chervonenkis (VC) dimension [37] of $\mathcal{H}$, denoted as d . For any $h \in \mathcal{H}$ and $\epsilon \in (0, 1)$, the following inequality holds with a probability of at least $1 - \epsilon$:

$$\begin{aligned} \theta^t(h) \leq & \theta^s(h) + \hat{d}_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^s, \mathcal{U}^t) \\ & + \lambda_{\mathcal{H}} + 4\sqrt{\frac{2d \log(2k) + \log\left(\frac{2}{\epsilon}\right)}{k}} \end{aligned} \quad (8)$$

where $\hat{d}_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^s, \mathcal{U}^t)$ denotes the estimated difference of $d_{\mathcal{H}\Delta\mathcal{H}}(P_X^s, P_X^t)$ across two distinct domains based on limited data samples. DA error bound (nonasymptotic) delineates a concept wherein learning algorithms are tailored to generalize toward a target domain utilizing a finite set of samples derived from both the source domains and target domains during the training phase. This error bound quantitatively measures the variance in risk between the target domain and source domain, encapsulating the difference in average loss. Specifically, it quantifies the distance between the average loss in the target and source domains. The bound is nonasymptotic, which means that it is valid for any size of the sample set and thus enables effective control of the generalization performance of the algorithm in practical applications.

In DA, the discrepancy between the source domain distribution P_X^s and the target domain distribution P_X^t poses a significant challenge. The distribution difference in the above error bound, denoted as $d(P_X^s, P_X^t)$, can impede the generalization capability of models trained on the source domain when applied to the target domain. Nonetheless, although direct control over the distribution difference can

be demanding, a potential approach is to train a representation function $g : \mathcal{X} \rightarrow \mathcal{Z}$ that maps the input data x to a representation space $\mathcal{Z}$, aiming to minimize the discrepancy between the distribution of representations from the two domains. Consequently, this approach aims to align the representation distributions of both domains more closely, thereby enhancing model generalization across domains. This technique, known as *DA based on domain-invariant representation* (DA-DIR), focuses on identifying a representation space $\mathcal{Z}$ where the distributions P_Z^s and P_Z^t from the source domain and target domain, respectively, are minimized. Achieving this involves employing methods such as adversarial training, discrepancy minimization, domain confusion loss, and transfer learning.

DA emerges as a formidable technique for enhancing ML model performance in instances where the source domain and target domain exhibit divergent distributions, a prevalent challenge across numerous real-world contexts.

C. Domain Generalization

DA operates under the premise that either labeled or unlabeled data from the target domain are available for model adaptation; this condition is not always met in real-world scenarios. Acquiring labeled data from the target domain can be prohibitively time-consuming, costly, or unfeasible due to ethical or legal restrictions. Moreover, the target domain's data distribution might remain elusive, complicating the application of conventional DA methods that depend on understanding this distribution. To tackle the issue, DG has been identified as a potent strategy to enhance the generalization of ML models across novel and unseen domains. This is achieved by training models on a heterogeneous collection of datasets encompassing various domains. The DG methodology is depicted on the left side of Fig. 5 and is concisely defined in Definition 3.

Definition 3 (DG): Given M source domains $\mathcal{S}_{\text{source}} = \{\mathcal{S}^i | i = 1, 2, \dots, M\}$, where $\mathcal{S}^i = \{(x_j^i, y_j^i)\}_{j=1}^{n_i}$ represents the i th domain, it is observed that the joint distributions between each pair of domains differ ($P_{XY}^i \neq P_{XY}^j$, for $1 \leq i \neq j \leq M$). The objective of DG is to develop a robust and generalizable labeling function $h : \mathcal{X} \rightarrow \mathcal{Y}$ solely utilizing the M source domain data, which minimizes the prediction error on an unseen target domain $\mathcal{S}_{\text{target}}$, where $P_{XY}^{\text{target}} \neq P_{XY}^i$ for $i \in \{1, 2, \dots, M\}$

$$\min_h \mathbb{E}_{(x,y) \in \mathcal{S}_{\text{target}}} [\ell(h(x), y)] \quad (9)$$

where $\mathbb{E}$ denotes the expectation operator and $\ell(\cdot)$ represents the loss function used to disparity the difference between the predicted output $h(x)$ and the true output y . The ultimate aim is to identify the optimal function $h(\cdot)$ that reduces the expected loss on $\mathcal{S}_{\text{target}}$, thereby demonstrating the model's capacity to effectively generalize to novel and unseen domains.

In DG, the concept of an underlying hyperdistribution is pivotal for understanding the relationship between source

domains and target domains. The probability distribution represents the joint distribution over all possible (x, y) for both source domains and target domains. Specifically, assume that all potential target distributions adhere to a hyperdistribution $\mathcal{P}$: $P_{XY}^t \sim \mathcal{P}$, and the same is true for the source distributions: $P_{XY}^1, P_{XY}^2, \dots, P_{XY}^M \sim \mathcal{P}$. This framework facilitates generalization to any target domain, regardless of its familiarity, by training a classifier that integrates domain-specific information P_X into its input mechanism. Consequently, for a domain characterized by the distribution P_{XY} , predictions are structured as $y = h(P_X; x)$. For such a function $h(\cdot)$, its average risk across all conceivable target domains is determined by the subsequent formula

$$\psi(h) = \min_h \mathbb{E}_{P_{XY} \sim \mathcal{P}} \mathbb{E}_{(x,y) \sim P_{XY}} [\ell(h(P_X, x), y)]. \quad (10)$$

However, it is often infeasible to evaluate expectations exactly. Instead, these expectations can be approximated using finite domains/distributions adhering to a specified probability distribution $\mathcal{P}$, along with finite samples of (x, y) drawn from each distribution. Given that $P_{XY}^1, P_{XY}^2, \dots, P_{XY}^M \sim \mathcal{P}$, we can use source domains and their corresponding supervised data to estimate the expectation. This is because the source domains and their associated data are assumed to be drawn from the same underlying distribution $\mathcal{P}$. As a result, the estimation of the average risk over all possible target domains based on the labeling function $h(\cdot)$ is as follows:

$$\hat{\psi}(h) = \frac{1}{M} \sum_{i=1}^M \frac{1}{n^i} \sum_{j=1}^{n^i} \ell(h(\mathcal{U}^i, x_j^i), y_j^i) \quad (11)$$

where M is the total number of domains, n^i is the number of samples in domain i , and the supervised dataset $\mathcal{U}^i = \{x_j^i | (x_j^i, y_j^i) \in \mathcal{S}^i\}$ represents the empirical estimation for the marginal distribution P_X^i of the features in domain i .

Theorem 2 (DG Error Bound [27], [38]): Let $\gamma := \min_{\pi \in \Delta_M} d_{\mathcal{H}}(P_X^t, \sum_{i=1}^M \pi_i P_X^i)$ with minimizer π^* represent the minimum distance between the target domain distribution P_X^t and the convex hull Λ formed by the source domain distributions P_X^i . Δ_M represents the $(M - 1)$ -dimensional simplex that approximates the target domain distribution P_X^t with a convex hull Λ of source domain distributions P_X^i . The minimizer π^* corresponds to the weights that give the best approximation $P_X^* = \sum_{i=1}^M \pi_i^* P_X^i$ of P_X^t within Λ , where π denotes a normalized mixing weights. The parameter $\rho = \sup_{P_X', P_X'' \in \Lambda} d_{\mathcal{H}}(P_X', P_X'')$ represents the diameter of the convex hull Λ . Then, the DG error bound can be expressed as

$$\theta^t(h) \leq \sum_{i=1}^M \pi_i^* \theta^i(h) + \frac{\gamma + \rho}{2} + \lambda_{\mathcal{H}, (P_X^t, P_X^*)} \quad (12)$$

where $\lambda_{\mathcal{H}, (P_X^t, P_X^*)}$ represents the ideal joint risk across the target domain P_X^t and the domain with the best approximator distribution P_X^* . $(M - 1)$ -dimensional simplex is often used to represent the mixing weights of source domains for approximating the target domain distribution. Specifically, for M source domain distributions, they can be viewed as M points in Euclidean space, and these points can be connected to form an $(M - 1)$ -dimensional simplex. In this simplex, each point corresponds to a source domain distribution, and each edge corresponds to a mixing weight of a source domain distribution (i.e., a real number between 0 and 1). The sum of mixing weights for each point is 1, representing their convex combination. Therefore, these mixing weights can be used to approximate the target domain distributions, thus achieving the purpose of DG. In practical implementation, some optimization algorithms are usually needed to determine the values of these mixing weights so that the approximated target domain distributions can be as close as possible to the actual one.

Similar to DA, the theoretical framework for DG emphasizes the importance of leveraging DIR techniques. These strategies aim to achieve two primary objectives: first, to minimize the risk across all source domains, as indicated by the initial term of the theoretical bound; and second, to diminish the discrepancies in the representation distribution between the source and target domains. The latter objective is quantitatively assessed using metrics γ and ρ , which serve to measure the divergence between the representations of source and target domains. By effectively minimizing this error bound, the model is trained to excel in generalizing across diverse domains, thereby ensuring robust performance on unseen domains.

To sum up, DG is a promising approach for developing models that adeptly navigate the challenges posed by distribution differences across various source domains. It achieves this by narrowing the distribution gap between source domains and target domains and fostering DIRs. This approach ensures that the models are well-equipped to excel in real-world scenarios, particularly when deployed on previously unseen target domains.

D. Federated Domain Generalization

Recently, the proliferation of IoT devices has created enormous amounts of diverse data from multiple domains, providing ideal conditions for DG research. The fundamental principle of DG is to access the multisource distributions in the learning process. However, the sensitive data generated by IoT devices, such as healthcare and transportation data, are often subject to strict privacy regulations, making it difficult to implement traditional DG methods that rely on centralized data processing. To tackle the issue of training models on data from multiple domains and obtaining a robust and generalizable predictive function while preserving data privacy, FDG has emerged as a promising solution [18]. FDG has many applications in areas such as healthcare, finance, and autonomous vehicles, where the data are often distributed across multiple devices or

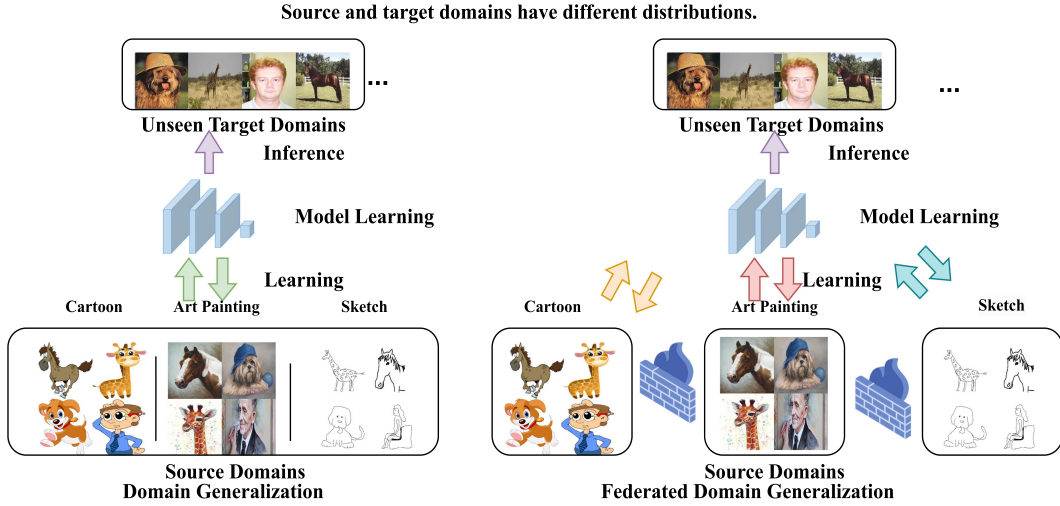


Fig. 5. Comparison of DG task (left) and FDG task (right). In DG, a centralized model is trained on multiple source domains and evaluated on unseen target domains with different distributions. In FDG, model training occurs collaboratively across decentralized clients holding heterogeneous source domains, without sharing raw data, to generalize to unseen target domains.

servers, and the privacy of the data is a critical concern. The process of the FDG task is illustrated on the right of Fig. 5 and summarized in Definition 4.

Definition 4 (FDG): Given M source domains $\mathcal{S}_{\text{source}} = \{\mathcal{S}^i | i = 1, 2, \dots, M\}$ involved in FL, where $\mathcal{S}^i = \{(x_j^i, y_j^i)\}_{j=1}^{n_i}$ denotes the i th domain, the joint distributions between each pair of domains are different: $P_{XY}^i \neq P_{XY}^{\hat{i}}, 1 \leq i \neq \hat{i} \leq M$. The objective of FDG is to develop a robust and generalizable labeling function $h : \mathcal{X} \rightarrow \mathcal{Y}$ using only M source domains data that can minimize the prediction error on unseen target domains $\mathcal{S}_{\text{target}}$ [where $P_{XY}^{\text{target}} \neq P_{XY}^i$ for $i \in \{1, 2, \dots, M\}$], while each domain i can only access its local data $\mathcal{S}^i$

$$\min_h \mathbb{E}_{(x,y) \in \mathcal{S}_{\text{target}}} [l(h(x), y)] \quad (13)$$

where $\mathbb{E}$ denotes the expectation operator and $l(\cdot)$ represents the loss function used to quantify the discrepancy between the predicted output $h(x)$ and the true output y . The primary goal is to identify an optimal function $h(\cdot)$ that minimizes the expected loss on $\mathcal{S}_{\text{target}}$, thereby indicating the model's proficiency in generalizing effectively to unseen domains.

It is crucial to acknowledge that the distinctions among these concepts may overlap, as illustrated in Fig. 6. The specific techniques and approaches used in each area can vary, but they all revolve around adapting or generalizing models to different domains or decentralized data settings.

III. METHODOLOGIES: A SURVEY

FDG methods have surfaced as a compelling approach to tackle domain shift and heterogeneity issues in FL, simultaneously ensuring data privacy. Although its potential to enhance model generalization and robustness in federated settings has attracted attention in recent years, research

in this area remains relatively sparse. In this section, we aim to fill this gap by providing an extensive review of the current literature on FDG, categorizing it into distinct groups according to their methodologies and motivations.

A. FDAL

FDAL [39] is of utmost importance in FDG as it enhances model generalization across multiple domains, promotes data efficiency, preserves data privacy, and enables the applicability of FL in real-world scenarios. The primary motivation of FDAL is to bolster the generalization potential of models across various domains. By aligning the domains and harmonizing their data representations, FDAL strives to mitigate domain shifts and enhance the performance of models on unseen domains. This aspect is crucial in practical applications, especially where data privacy and security are of utmost importance. FDAL facilitates efficient knowledge transfer while maintaining the

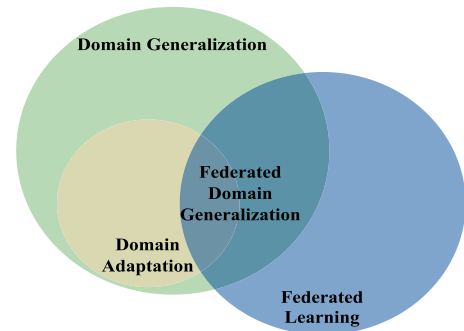


Fig. 6. Conceptual relationships among FL, DA, DG, and FDG. The figure illustrates the conceptual overlaps and distinctions between the four paradigms.

decentralized and secure nature of sensitive data. The objective of alignment-based FDG methods, focusing on joint distribution, is articulated as follows:

$$\begin{aligned} P(X, Y) &= \sum_{i=1}^M P(D_i) P(X|D_i) P(Y|X, D_i) \\ &= \sum_{i=1}^M P(D_i) P(Y|D_i) P(X|Y, D_i) \end{aligned} \quad (14)$$

where $P(X, Y)$ represents the joint distribution of input features X and corresponding labels Y across all domains; $P(D_i)$ represents the distribution of domain i , indicating the probability of selecting samples from domain i ; $P(X|D_i)$ represents the conditional distribution of input features X given domain i , capturing the domain-specific characteristics of the data; and $P(Y|X, D_i)$ represents the conditional distribution of labels Y given input features X and domain i , describing the relationship between the features and labels in each domain.

$P(X, Y)$ sums over all domains i to consider the contribution of each domain to the overall joint distribution $P(X, Y)$. The two equations represent two different decompositions of the joint distribution, highlighting different relationships between the variables. In FDG, our main focus is on aligning the marginal distributions of the source domains. Specifically, we aim to align $P(D_i)P(X|D_i)$ to reduce the distribution shift between domains. This is because we assume that the posterior probability $P(Y|X, D_i)$ remains relatively stable and has a smaller impact on the model. Therefore, domain alignment methods primarily concentrate on adjusting the distribution of input features to mitigate differences across domains and improve the model's generalization performance on unseen domains.

1) *Adversarial Feature Alignment*: Adversarial feature alignment involves techniques that leverage adversarial learning to align feature distributions across different domains, thereby reducing domain discrepancy and enhancing model generalizability. In FDG, adversarial feature alignment plays a critical role in generating models that are not only accurate but also robust to variations in data distribution, ultimately leading to improved performance on unseen data.

Peng et al. [40] proposed federated adversarial alignment to align the feature distributions between different domains in FL by dividing optimization into two independent steps: training domain-specific local feature extractors and training a global discriminator. The following steps are as follows.

- 1) *Local Feature Extractors*: A feature extractor G_s is trained for the source domain $\mathcal{S}^i$, and a feature extractor G_t is trained for the target domain $\mathcal{S}^t$.
- 2) *Adversarial Domain Alignment*: For each source–target domain pair $(\mathcal{S}_s, \mathcal{S}_t)$, an adversarial domain identifier

DI is trained to align the feature distributions of the two domains adversarially. Initially, the DI is fine-tuned to precisely identify the originating domain of the features. Following this, the generator models (G_s and G_t) are trained to confuse DI by producing features that make it tough for DI to differentiate between the source and target domains. The objective of DI is defined as follows:

$$\begin{aligned} L_{\text{advDI}_i}(\mathcal{S}_s, \mathcal{S}_t, G_s, G_t) &= -\mathbb{E}_{s_s \sim \mathcal{S}_s} [\log \text{DI}_i(G_s(s_s))] \\ &\quad - \mathbb{E}_{s_t \sim \mathcal{S}_t} [\log (1 - \text{DI}_i(G_t(s_t)))]. \end{aligned} \quad (15)$$

In the second step, the objective for updating the generators G_s and G_t (denoted as L_{adv_G}), while keeping L_{advDI_i} unchanged, is defined as follows:

$$\begin{aligned} L_{\text{adv}_G}(\mathcal{S}_s, \mathcal{S}_t, \text{DI}_i) &= -\mathbb{E}_{s_s \sim \mathcal{S}_s} [\log \text{DI}_i(G_s(s_s))] \\ &\quad - \mathbb{E}_{s_t \sim \mathcal{S}_t} [\log \text{DI}_i(G_t(s_t))]. \end{aligned} \quad (16)$$

Wang et al. [41] developed a multiclient feature alignment framework employing adversarial learning, wherein a GAN leverages class-wise information to generate adaptive reference distributions. This methodology, aimed at extracting DG features across diverse clients, effectively reduces distribution discrepancies. The training process is described as follows.

- 1) *Feature Extraction and Generation*: A feature extractor $F(\cdot)$ extracts real features by generating a representation $f^{s,i} = F(x^{s,i})$ for each source client i , where $x^{s,i}$ is the input data. Following this, a generator $G(\cdot)$ employs random noise z and a one-hot vector y to produce fake features $f_{\text{fake}}^{s,i} = G(z^{s,i}|y)$, maintaining consistency across clients by adhering to a shared, predefined distribution $p(f_{\text{fake}})$.
- 2) *Discriminator Update Process*: The discriminator $D(\cdot)$ is refined through the integration of both real and fake features, where real features serve as negative samples and fake ones as positive, employing a least-squared estimation approach within its loss function $L_{\text{adv}_d}^i$ to bolster convergence and discriminative proficiency between feature types

$$\begin{aligned} L_{\text{adv}_d}^i &= \mathbb{E}_{z \sim p(f_{\text{fake}}^{s,i})} [D(f_{\text{fake}}^{s,i}|y)]^2 \\ &\quad - \mathbb{E}_{x^{s,i} \sim p(f^{s,i})} [(1 - D(f^{s,i}|y))]^2. \end{aligned} \quad (17)$$

- 3) *Generator and Feature Extractor Update Process*: With the discriminator's parameters fixed post-training, the feature extractor and generator are updated through an adversarial process. The feature extractor aims to produce features that the discriminator perceives as

positive (real), whereas the generator strives to create fake features that are indistinguishable from real ones to the discriminator. This adversarial interplay is captured in the loss functions $L_{adv_f}^i$ and $L_{adv_g}^i$ for the feature extractor and generator, respectively,

$$L_{adv_f}^i = \mathbb{E}_{x^{s,i} \sim p(f^{s,i})} \left[(1 - D(f^{s,i}|y))^2 \right] \quad (18)$$

$$L_{adv_g}^i = \mathbb{E}_{z \sim p(f_{fake}^{s,i})} \left[D(f_{fake}^{s,i}|y)^2 \right]. \quad (19)$$

- 4) *Iterative Optimization*: The training process iteratively optimizes the discriminator and generator until reaching a Nash equilibrium, indicative of the discriminator's diminished ability to distinguish between real and synthetic features.

While both the above methodologies share the overarching goal of leveraging adversarial learning for DG in FL, they differ in their specific techniques and the scope of their application. The former concentrates on domain-specific alignment, whereas the latter pursues a more generalized strategy for feature alignment across multiple clients.

2) *Minimizing Contrastive Loss*: The contrastive loss [7], [42] encourages similar representations for samples from the same domain and dissimilar representations for samples from different domains by maximizing agreement within positive pairs and minimizing agreement within negative pairs. The essence of minimizing contrastive loss is to sculpt a representation space where akin samples are drawn nearer, while disparate samples are distanced. The formulation of contrastive loss is delineated as follows:

$$L_{\text{Contrastive}} = \frac{1}{2N} \sum_{i=1}^N y d^2 + (1 - y) \max(\text{margin} - d, 0)^2 \quad (20)$$

where N is the number of sample pairs, y is a binary label indicating whether a pair of samples are similar (positive pair, $y = 1$) or dissimilar (negative pair, $y = 0$), d is the distance between the representations of the paired samples, and margin is a hyperparameter that defines the minimum desired distance between representations of dissimilar samples.

Liu et al. [43] introduced a federated contrastive learning approach that integrated contrastive learning with FL to enhance model generalization across heterogeneous medical data, demonstrating superior performance in medical image classification. To enhance model invariance across domains, an alignment loss is deployed between the original sample and its hallucinated counterpart using a negative-free contrastive loss [44] at the logit level.

Yu et al. [45] presented a contrastive-enhanced DG framework within FL, which integrates an advanced instance normalization module and a prototype-based contrastive loss to improve model generalization and

discrimination while preserving data privacy. The prototype-based contrastive loss for client k , denoted as l_{con}^k , is defined as follows:

$$l_{\text{con}}^k = \frac{1}{N_k} \sum_{i=1}^{N_k} \sum_{c=1}^C \mathbb{1}(y_k^i = c) \left(-\log \frac{S_i^+}{S_i^+ + S_i^-} \right), \text{ with} \quad (21)$$

$$\begin{cases} S_i^+ = e^{\text{sim}(z_k^i, z_{\text{global},c})/\tau} \\ S_i^- = \sum_{c' \leq C, c' \neq c} e^{\text{sim}(z_k^i, z_{\text{global},c'})/\tau} \end{cases}$$

where N_k denotes the number of data points for client k and C represents the total number of classes. The S_i^+ measures the similarity between z_k^i and the global prototype of the same class, while S_i^- measures the sum of similarities between z_k^i and the global prototypes of all other classes. The function $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity, and τ is a temperature parameter that scales the similarities.

Heterogeneous FL primarily addresses challenges such as statistical, model, communication, and device heterogeneity to enhance the average performance of models across all clients while ensuring data privacy [46]. In contrast, FDG focuses on training models across multiple domains with varying distributions, aiming to enable the model to generalize effectively to previously unseen domains. Despite their distinct objectives, the techniques developed for heterogeneous FL can also be applied to FDG, as both fields involve addressing distributional discrepancies. For example, Li et al. [47] proposed the MOON framework that leverages model-level contrastive learning to bridge the gap between global and local features and improves the performance of FL. To tackle the underexplored issues of data heterogeneity and class imbalance in FL, a novel privacy-preserving framework, FedIIC [48], has been proposed. This framework adeptly calibrates deep models to tackle imbalanced training through the use of intraclient and interclient contrastive learning techniques, resulting in superior performance across both realistic and established scenarios. Tan et al. [49] proposed a lightweight framework called FedPCL, enabling clients to jointly learn by fusing representations from multiple pre-trained models, improving each client's ability to leverage class-relevant information in a personalized manner while maintaining compact shared knowledge for efficient communication. Qiao et al. [50] presented FedCCL, a novel FL framework that integrates local and global clustered feature contrast strategies to address intradomain and interdomain heterogeneity challenges, enhancing the generalization of local models.

While methods such as MOON, FedIIC, FedPCL, and FedCCL primarily address client heterogeneity by minimizing contrastive loss, this approach also aids in FDG. By learning invariant representations across clients, these methods enhance the model's generalization to new domains, akin to unseen clients in FL settings. For instance, as an extension of MOON [47], an improved contrastive loss [51] is utilized by considering the projected features of

pseudo-data as positive pairs and incorporating the projected local features of both pseudo-data and local data as negative pairs. Moreover, FedPCL [49] has been shown to significantly improve generalization in FL scenarios where clients have non-IID data, demonstrating its utility in learning domain-invariant features. Additionally, Fed-CCL focuses on capturing intraclass information through local clustering while simultaneously promoting cross-client knowledge sharing, enhancing the model's ability to generalize across diverse data distributions.

Minimizing contrastive loss enhances FDG by aligning representations across different domains, effectively reducing domain shift and boosting the model's generalization capabilities across diverse domains. Additionally, addressing client heterogeneity through contrastive loss is a step toward achieving robust FDG.

3) *Wasserstein Distance*: The WD, also known as the Earth mover's distance, is a measure of the dissimilarity between probability distributions. In FDG, the WD serves as a quantitative measure to evaluate the disparity between distributions across various domains. This approach facilitates the development of models with enhanced generalizability to previously unseen domains.

Chen et al. [52] introduced a Wasserstein-based feature critic with meta-optimization into FL, which effectively improved the robustness and generalization capability of the proposed method. Nguyen et al. [53] proposed a Wasserstein distributionally robust optimization scheme called WAFL for FL, which is more general than related approaches and is robust to all adversarial distributions inside the Wasserstein ball. Lin et al. [54] introduced a robust federated meta-learning framework that utilizes WD and Lagrangian relaxation to efficiently optimize the distance metric, enabling gradient-based methods to solve the problem effectively.

In summary, the integration of the WD within the objective function of FDG methods facilitates the learning of DIRs, enabling improved generalization across federated domains.

4) *Minimizing MMD*: MMD [55] serves as a statistical metric for quantifying the differences in probability distributions across various domains. In FDG, MMD is employed to reduce the distributional shifts between domains, thereby facilitating domain alignment. The objective is to minimize MMD to foster DIRs, enabling the model to capture and leverage commonalities across domains while retaining unique domain features. The calculation of MMD is as follows:

$$\text{MMD}^2(P, Q) = \left\| \frac{1}{n'} \sum_{j=1}^{n'} v(x_j) - \frac{1}{n''} \sum_{j=1}^{n''} v(y_j) \right\|^2 \quad (22)$$

where P and Q are the probability distributions of the feature representations of two domains, n' and n'' are

the number of samples in each domain, and $v(\cdot)$ is a feature mapping function that maps the samples to a high-dimensional space.

Chen et al. [56] implemented D-WFA using the MMD distance between the source and target clients, which helps to reduce domain differences and improve the generalization capability. To sum up, we can effectively solve the domain shift problem and enhance the generalization capabilities of FL models by incorporating MMD in FDG. The proposed method [57] enables the aggregated gradient to incorporate information from multiple domains by connecting gradient alignment with MMD, aiming for better generalization on unseen domains. Zhang and Li [58] employed MMD to measure and optimize the distance between different domains by calculating the squared distance between the kernel embeddings of the data representation distributions within the reproducing kernel Hilbert space, with unbiased empirical estimation used to reduce computation load. Zhang et al. [59] introduced a blockchain-based decentralized FTL approach for collaborative machinery fault diagnosis. This method overcomes the issues of data quality and quantity by guaranteeing data security and privacy, thereby achieving high testing accuracies and presenting a viable solution for real-world industrial applications. Wang et al. [41] developed a federated adversarial DG network for machinery fault diagnosis, facilitating collaborative training between a central server and multiple clients to ensure data privacy. This approach resulted in significant performance improvements and showed potential for tasks related to data privacy-preserving generalization in diagnostics.

Overall, minimizing the MMD facilitates FDG by encouraging feature representations from various domains to exhibit similar means in high-dimensional space, thereby reducing distributional discrepancies. This promotes the alignment of domains and improves the generalization capability of models across different domains.

5) *Prediction Assessment*: Prediction consistency and prediction disagreement represent two interconnected aspects of model evaluation in FDG, each playing a distinct role in enhancing the generalization capabilities of FL systems across decentralized domains.

Prediction consistency refers to the similarity in predictions made by different models trained across various clients or by a single model under different conditions for the same input. In FDG, consistency is indicative of a model's ability to learn domain-invariant features that are robust across unseen domains. High prediction consistency suggests that models are capturing the underlying patterns in the data that are common across domains, thereby improving the model's generalization to new and unseen domains. Zhang and Li [60] leveraged prediction consistency to enhance the generalization capability of the model. The inference results for the target domains are obtained by majority voting on the predictions made by multiple source domains. Considering the fact that

these source domains have different parameters, prediction consistency generally improves the model's generalization capability while reducing the risk of overfitting. Thus, it is desirable to minimize variations in the inference results among the source domains. To achieve this, the loss function L_p is defined as follows:

$$\min L_p = \frac{1}{n^t M} \sum_{j=1}^{n^t} \sum_{i=1}^M \|I^{s,i}(F^t(x_j^t)) - \bar{I}(F^t(x_j^t))\|$$

$$\bar{I}(F^t(x_j^t)) = \frac{1}{M} \sum_{i=1}^M I^{s,i}(F^t(x_j^t)) \quad (23)$$

where n^t represents the number of target client data samples, M represents the number of source inference models, $F^t(x_j^t)$ represents the feature representation of the target domain data x_j^t in the model, $I^{s,i}$ represents the inference module at the i th source domain, and $\bar{I}(F^t(x_j^t))$ denotes the mean value of predictions obtained from the source inference models.

Prediction disagreement underscores the variations in outcomes generated by distinct models or a single model under varying conditions for the same input, revealing critical insights into the diversity and domain-specific characteristics of decentralized datasets. This divergence is instrumental in fostering the development of models with enhanced generalizability and robustness within FDG, thereby ensuring the aggregated model's efficacy across diverse unseen domains. Guo et al. [61] introduced FEDIIR, a novel approach that implicitly learns invariant relationships for OOD generalization in FL, leveraging prediction disagreement and interclient gradient alignment to enhance model performance while adhering to privacy and communication efficiency principles.

While prediction consistency aims to ensure that the model learns generalizable features, prediction disagreement offers insights into model diversity and domain-specific biases. Together, they provide a comprehensive framework for evaluating and improving the generalization performance of federated models. By judiciously analyzing and leveraging both consistency and disagreement, FDG can effectively address the challenges posed by domain drift and data decentralization, leading to models that are both robust and highly generalizable across diverse domains.

6) *Prototype Learning*: Prototype learning, a pivotal concept in ML, emphasizes identifying “prototypes”—representative samples within a dataset that encapsulates the diversity present across the data. In FDG, prototype learning entails each client pinpointing prototypes specific to their domain, while a global model synthesizes these data to discern shared prototypes, thereby uncovering commonalities across domains and concurrently safeguarding data privacy.

Huang et al. [62] introduced federated prototypes learning for addressing domain shift in FL by constructing cluster and unbiased prototypes to provide domain knowledge and a fair convergence target, demonstrating its effectiveness and efficiency on tasks. Liu et al. [63] introduced PGCT, a novel prototype-guided cross-training mechanism for FL, designed to address data heterogeneity and knowledge forgetting by leveraging client-specific data prototypes for consistent representation learning and feature augmentation, demonstrating superior performance over existing methods. Yu et al. [45] introduced a prototype-based contrastive loss to enhance class-wise alignment within the embedding space, thereby improving the model's generalization capabilities. Li et al. [64] proposed the FedCPG framework, which utilizes a model decoupling approach to minimize challenges associated with device diversity. Additionally, it incorporates class prototype constraints to guide local models, ensuring that the global model can reflect the common characteristics of all domains and enhancing the model's generalization capabilities.

Leveraging prototype learning within FDG adeptly mitigates the issues posed by domain shift and privacy concerns. This approach enables the crafting of models that are not only adept at preserving privacy and efficiency but also excel in generalizing to novel, unseen domains.

B. Data Manipulation

The performance of ML models is significantly influenced by the quality and quantity of the training data, especially in DG where the models are expected to generalize well to unseen domains. However, acquiring a large and diverse set of training data that covers all possible domains is often impractical or costly. In addition, the privacy concerns associated with sensitive data hinder centralized data collection and storage. In FDG, DM techniques have been devised to surmount the aforementioned difficulties, thereby generating more diverse and representative training datasets. One such technique is data augmentation, which can be used to create additional training data by modifying the existing data through techniques such as rotation, scaling, and adding noise. Furthermore, adversarial data augmentation can be used to generate difficult examples that can help improve the model's robustness against domain shifts and adversarial attacks.

The objective of DM can be expressed mathematically as

$$\min_{\omega} \sum_{i=1}^M \nu_i \mathcal{L}_i(\hat{S}^i, \omega) + \xi R(\omega) \quad (24)$$

where M is the total number of domains, ω represents the model parameters to be learned, ν_i represents the importance of domain i in FL and is determined based on the quantity and quality of the domain i , and $\mathcal{L}_i(\hat{S}^i, \omega)$ measures the performance of the model on the domain i and includes both real and synthetic data generated by

data generation techniques. The hyperparameter ξ regulates the balance between fitting the manipulated data and model complexity, where a higher ξ favors simplicity and may increase training error, while a lower ξ permits greater complexity, enhancing fit but risking overfitting. The regularization term $R(\omega)$ encourages the model to learn domain-invariant features that are less sensitive to domain shifts and can improve generalization performance. Depending on the specific techniques used and the formula for manipulating the data $\hat{S}^i$, existing work on DM can be broadly categorized into two main categories: data augmentation and data generation.

1) *Data Augmentation-Based FDG*: It involves applying various transformations to existing training data, such as rotating, scaling, or adding noise, to create new training data that are similar to the original ones. This can help increase the diversity and robustness of the training data and improve the model's generalization performance to unseen domains. Here, $\hat{S}^i = S^i \cup A^i$, where S^i is the local real dataset of domain i and A^i is the set of random perturbations used for data augmentation.

a) *Domain randomization*: In FDG, domain randomization can be used to simulate the variability of the data across different clients by randomizing certain aspects of the data during training. This can help the model learn to be robust to domain shifts and better generalize to unseen clients.

Duan et al. [65] used local domain randomization augmentation to create a globally balanced distribution. Guo et al. [51] proposed FedDebias, a unified algorithm that leverages mean-based random noise augmentation that can reduce the learning bias on local features and augment model efficacy without introducing real data. Liu et al. [18] employed the data augmentation of random rotation, scaling, and flipping in FL tasks. Atwany and Yaqub [66] applied data augmentations including resizing, horizontal and vertical flipping, random grayscale, ColorJitter, random rotation, translation, and Gaussian blur to fundus images to improve model performance and intradomain generalizability.

b) *Style transfer data augmentation*: Style transfer data augmentation [67] is a technique that leverages the principles of neural style transfer to generate new training examples by combining the content of one image with the style of another, enhancing the diversity and richness of the training dataset.

Shenaj et al. [68] took full advantage of the source dataset during the pretraining stage with style transfer data augmentation, randomly loading the target styles in the source images to mimic the target distributions. Chen et al. [69] proposed cross-client style transfer (CCST) in FL involves each client computing and sharing their local styles with the central server, which can be either single-image styles or a local image style bank, to facilitate the data augmentation process and enhance diversity in the training data. Georgiadis et al. [70] introduced style

transfer FL aimed at COVID-19 image segmentation, tackling data variability and privacy constraints. This method utilizes a denoising CycleGAN at each client node to enhance robustness and attain performance on par with centrally trained models. Lewy et al. [71] developed an augmentation process that partitions images into batches within each node, computes statistics from a randomly chosen image, and then applies these statistics to augment each image in the batch, effectively emulating a node-specific style transfer augmentation. Yan et al. [72] presented FedRDN, an innovative and potent data augmentation approach designed to alleviate feature shifts in skewed feature distribution scenarios within FL. This approach showcases robust scalability and generalizability, providing a plug-and-play solution that seamlessly bolsters the efficacy of diverse FL models.

Apart from the data augmentation-based FDG mentioned above, there are other popular techniques for data augmentation. *Zero-shot augmentation* generates diverse and realistic augmented samples by leveraging external knowledge or pretrained models without requiring any additional labeled data, enhancing model generalization and performance. Hao et al. [73] introduced Fed-ZDA, an FL system that leverages zero-shot data augmentation to generate pseudo-exemplars of unseen classes while ensuring privacy, aiming to enhance fairness and accuracy performance uniformity across clients in federated networks. *Semantic knowledge augmentation* is a technique used in ML to improve model performance and generalization by leveraging external semantic knowledge to generate more accurate, diverse, and semantically consistent training data. Sun et al. [74] enrich the semantic space by combining text embedding, Gaussian noise, and attribute labels during the training process, making the model better adapt to the instance-level visual space and enhancing performance and generalization capabilities. *Deep stacked data augmentation*-based FDG leverages the power of DL, data augmentation, and FL techniques to enhance the generalization capability of models across all domains. By augmenting the data and aligning the gradients, the models become more robust and capable of handling unseen domains, thereby improving the performance and applicability of FL in real-world scenarios. Xing Tian et al. [57] proposed a deep stacked transformation-based data augmentation method, referred to as BigAug, to enhance model generalization to unseen domains. The approach demonstrated strong generalization performance across multiple unseen datasets. Yu et al. [75] proposed a novel data augmentation method, termed symmetry-inspired data augmentation, which notably expands the sample size while minimizing memory requirements, thus significantly advancing FDG within the realm of medical imaging. These methods have proven effective in creating diverse, domain-invariant training samples, thereby enabling efficient knowledge transfer and adaptation across disparate domains, all the while maintaining data privacy under the FL paradigm.

2) *Data Generation-Based FDG*: It involves synthesizing new training samples from scratch using generative models such as variational autoencoders or GANs. These models are designed to generate samples that balance statistical similarity and content diversity. Statistical similarity ensures that the generated data remain representative of the source domain, thereby facilitating effective learning, while content diversity broadens the model's exposure to potential shifts in unseen target domains. This, in turn, enhances the model's generalization capability. By addressing the limitations posed by scarce or biased training data, data generation-based FDG contributes to more robust and adaptable model performance in previously unseen domains. Here, $\hat{S}^i = S^i \cup C^i$, where S^i is the local real dataset of domain i and C^i is the virtual domain dataset used for data generation

a) *Generative model*: The generative model can be used to generate synthetic data that capture the common characteristics of the different domains, enabling better DG. The choice of generative model for FDG depends on factors such as the characteristics of the data from different domains, the availability of labeled data in each domain, and the desired level of DG.

i) *GAN-based FDG*: Zhang and Jiang [76] proposed FedDA, a data augmentation method based on GAN for generating high-quality overlap samples by learning features from both overlap and nonoverlap data, effectively expanding the available training data and enhancing the effect of data augmentation. Guo et al. [77] demonstrated the efficacy of ACGAN in managing data imbalance and enhancement tasks. Leveraging its mechanism, ACGAN presented a viable data conduit for preserving data privacy in FTL, marking a novel application for ACGAN. To enable membership inference on other participants, Chen et al. [78] proposed a novel approach that utilizes locally deployed GANs to generate samples with all labels, thereby obtaining the data distribution necessary for the inference process. Zhang et al. [79] leveraged GANs to generate synthetic data with the same underlying distribution as the original dataset, guided by the target FL model's discriminator, thereby enriching the training dataset and improving the imitative capability of the generated images. Zhang et al. [80] proposed a DPGAN framework that allows different hospitals to employ a privacy-preserving data augmentation method by utilizing distributed DPGAN models to generate high-quality training samples, addressing the issue of insufficient training data and subsequently achieving accurate detection using the ResNet model in FL.

Yan et al. [81] synthesized samples by a privacy-preserving GAN, which solves the cross-client variation problem and protects privacy with wide applicability. Li et al. [82] proposed an improved FL-GAN to learn globally shared GAN models by aggregating locally trained generator updates with the maximum mean difference. Sariyildiz and Cinbis [83] built a generative model to generate training samples for unseen classes upon the WGAN [84], [85] that takes a combination of noise vectors

and class embeddings as input, enabling the production of class-specific samples based on the information provided by the class embedding. Tang et al. [86] introduced virtual homogeneity learning, a method for handling data heterogeneity in FL, which generates a separable and homogeneous virtual dataset by utilizing StyleGAN [87] from shared noise across clients to improve convergence speed and generalization performance while preserving privacy. Zhou and Konukoglu [88] developed FEDFA, an innovative FL algorithm that mitigates feature shift among clients by probabilistically augmenting local feature statistics based on global information from the entire federation, offering a robust solution to improve model performance in the presence of non-IID data distributions.

ii) *Diffusion model-based FDG*: In contrast to GANs, diffusion models have emerged as a promising alternative for FDG, offering improved stability and generative quality. Recent studies have highlighted their potential. Li et al. [89] proposed FedDiff, a novel multimodal collaborative diffusion-based FL framework that integrates dual-branch diffusion models with a lightweight communication module, enabling secure and efficient fusion of heterogeneous data across clients. This framework achieves state-of-the-art performance and robust privacy protection in multimodal classification tasks. Wang et al. [90] introduced DACOA, a diffusion-based, text-guided style transfer method for FL that enhances DG by generating high-quality cross-domain augmentations and aligning both features and semantics. Liang et al. [91] developed a data replay approach based on pretrained conditional diffusion models. By incorporating contrastive learning, this method enhances DG in federated continual learning, mitigates catastrophic forgetting, and generates high-quality data at low computational cost, outperforming established baselines.

b) *Mixup-based augmentation*: The models mentioned above can serve as starting points, and further customization or hybrid approaches may be necessary based on the specific requirements of the FL setting. Mixup-based augmentation is forced to learn features that are common across the clients by mixing samples from different clients during training, thereby improving its ability to generalize to new, unseen clients.

Shin et al. [92] introduced XorMixup, a privacy-preserving data augmentation technique based on XOR operations, designed to tackle the non-IID data hurdle in FL. This approach involves collecting encoded data samples from various devices and decoding them exclusively with each device's data, ensuring privacy and enhancing data diversity. Yoon et al. [93] proposed FedMix, a privacy-protected data augmentation technique, averaging local batches and subsequently applying Mixup in local iterations to generate augmented data. The proposed approach aims to protect the privacy of local data in FL scenarios while improving performance in non-IID settings. Yao et al. [94] generated a certain proxy to approximate the target distribution for aligning features from different domains

via mixup-based augmentation. Seunghan et al. [95] proposed a method for generating diverse domains by mixing local and global feature statistics while preserving data privacy. In this approach, MixIG is constructed by randomly interpolating instance-level and global statistics.

c) *Normalization techniques*: They unify the distribution of data across features or datasets, enhancing optimization efficiency and mitigating issues like gradient disappearance or amplification. Within FDG, these techniques are crucial for reconciling data heterogeneity, thus aiding in the development of consistent and generalizable attributes within robust global models.

Yu et al. [45] developed an enhanced instance normalization module designed to concentrate on task-relevant features while minimizing domain-specific information, thereby boosting the discriminative and generalization capabilities of the local model. Li et al. [96] introduced FedBN, leveraging local batch normalization to address feature shift, outperforms traditional FedAvg and advanced FedProx for non-IID data in extensive testing. Zhu et al. [97] developed a batch-instance style normalization block tailored for FL to address the domain gap attributed to stylistic variances. This batch-instance style normalization block is integrated with the segmentation backbone network, forming batch-instance style normalization network, which proficiently learns intradomain features and concurrently neutralizes interdomain style disparities, all without necessitating data access from other centers. Yuan et al. [16] employed hybrid batch-instance normalization and collaboration of frozen classifiers to improve the generalization of feature extractors. Yu et al. [75] proposed a masked adaptive instance normalization technique, addressing the challenge of generalizing models to unseen sites by minimizing intersite discrepancies and standardizing input images from different sites into a site-unrelated style.

By incorporating data generation techniques in FDG, models can benefit from increased diversity and representativeness of the training data. This helps in capturing the shared features and reducing the domain-specific differences, ultimately leading to improved generalization performance across domains.

C. Learning Strategies

In addition to DM, the field of FDG has garnered considerable attention within the broader ML paradigm. FDG research can be classified into several distinct approaches, including representation learning-based FDG, federated DA-based FDG, FTL-based FDG, federated adversarial learning-based FDG, style transfer-based FDG, federated meta-learning-based FDG, and other strategies. These approaches are devised to tackle the issue of training models that can effectively generalize to unseen domains by employing various techniques and principles.

1) *Representation Learning*: Representation learning [98] has long been a focal point in ML, playing a pivotal

role in the success of DG. Representation learning in FDG follows the same principles as representation learning in general but is tailored to the FL setting. The basic principle involves training a neural network to learn a mapping function that transforms the input data from each domain into a common representation space. The objective function used in representation learning for FDG typically includes components that encourage DIRs while preserving the utility of the learned representation for downstream tasks. The goal is to acquire DIRs that encapsulate common features across various federated domains, facilitating models to generalize effectively to unseen domains. In FDG, we decompose the prediction function w as $w = u \circ z$, where z represents the representation learning function and u represents the classifier function. Mathematically, it can be formulated as

$$\min_{u,z} \sum_{s \in \mathcal{S}} [\mathbb{E}_{(x,y) \sim P(s)} \mathbb{L}_t(u(z(x)), y) + \chi \mathbb{L}_d(z(x), D)] \quad (25)$$

where $\mathcal{S}$ represents the set of source domains and $\mathbb{E}_{(x,y) \sim P(s)}$ represents the expectation over the data samples (x, y) from a specific source domain $s \in \mathcal{S}$. $\mathbb{L}_t(\cdot)$ denotes the task-specific loss function, which measures the discrepancy between the predicted output of the representation function $u(z(x))$ and the true label y for a given input x . $\mathbb{L}_d(\cdot)$ represents the domain discrepancy loss function, which quantifies the difference between the representations $z(x)$ and the domain labels D . The domain discrepancy loss function can be defined using various approaches, such as MMD, ADA, or adversarial neural networks, depending on the specific method used for FDG. This loss encourages the learned representations to be domain-invariant, capturing shared knowledge across different source domains. The hyperparameter χ controls the tradeoff between the task loss and the domain discrepancy loss, determining the relative importance of task performance and domain invariance in the learned representations.

a) *DIR-based FDG*: It refers to the problem of learning models that can generalize well to unseen domains in the FL setting while also capturing DIRs.

i) *Federated adversarial learning*: Federated domain adversarial learning combines the concepts of FL and adversarial training to train a global model on multiple source domains. By incorporating a domain classifier, the model learns DIRs that generalize well across domains.

Xu et al. [44] outlined a novel federated adversarial domain hallucination learning framework for FDG, making the final model more robust to unseen domain shifts. Zhang et al. [99] introduced FedADG, employing federated adversarial learning to harmonize distributions across diverse source domains. This method yields a universal feature representation with strong generalization capabilities across unfamiliar target domains while safeguarding

local data privacy. Peng et al. [40] extended adversarial adaptation techniques to the constraints of the FL setting and leveraged feature disentanglement to bolster knowledge transfer. Zhang and Li [60] introduced deep adversarial networks as a solution to effectively bridge the gap between domain distributions while maintaining data privacy. Micaelli and Storkey [100] developed a training approach in which a student network learns to align its predictions with those of a teacher network, solely relying on an adversarial generator to discover images where the student exhibits poor alignment with the teacher and utilizing them for student training, without relying on any data or metadata. Zhang et al. [101] developed a knowledge-agnostic approach for on-device knowledge transfer by adversarially training a generative model with the global model to obtain higher accuracy and better generalization performance. Dalmaz et al. [102] introduced a novel specificity-preserving FL technique for MRI contrast translation. This approach leveraged an adversarial model to dynamically normalize feature maps across the generator based on site-specific latent variables. Kang et al. [103] introduced a fine-grained ADA approach aimed at minimizing feature dimensionality, improving model interpretability, and enabling the acquisition of domain-invariant features. Zhang et al. [104] developed a multihop graph pooling adversarial network tailored for cross-domain remaining useful life prediction, addressing the domain drift among varied clients by accommodating the diverse data distribution. Yu et al. [75] utilized a gradient reversal layer within the U-Net encoder to foster DIRs, bolstering model generalization. Ahmed et al. [105] presented FAME, a framework that combines FL and GANs to improve MRI reconstruction from undersampled data by training local generators across multiple institutions and using an adaptive aggregation strategy based on performance metrics and clinical relevance, ensuring privacy, data diversity, and improved model generalization. Although FAME adopts an adaptive aggregation strategy to handle data diversity, further refinement may be needed for extreme non-IID data distributions among different institutions.

In summary, federated adversarial learning contributes to improved generalization performance while ensuring privacy preservation and data security in the FL setting.

ii) Federated feature alignment: It, as utilized in FDG, addresses the challenge of domain shift across heterogeneous domains in FL. Its objective is to align feature representations of data samples from different domains, thereby enhancing the generalization capability of the federated model.

Nguyen et al. [106] presented FedSR, a novel representation learning framework for FL, which enables DG while preserving privacy and decentralization and incorporates conditional mutual information (CMI) and L2-norm regularizer (L2R) regularization techniques to improve generalization performance. To align features across two domains in FL without direct access to target data, Yao et al. [94]

proposed constructing a proxy by applying a reweighted mixup of server-side data, which effectively improves model generalization, robustness, and stability. Furthermore, additional sampling was conducted based on client data density using Gaussian mixture models to generate a refined proxy set. Zhang et al. [107] proposed the FedUFO method, and an adversary module is introduced to minimize feature divergence among different clients, while consensus losses were employed to mitigate inconsistencies in optimization objectives, enabling unified feature learning and alignment in non-IID FL. Sun et al. [108] proposed FedKA, a novel FDG method that utilizes feature distribution matching and a federated voting mechanism to align domain-specific features and generate pseudo-labels for global model fine-tuning, enabling domain-invariant learning in the presence of unknown client data. To learn feature representations from raw input data, Kang et al. [109] utilized two neural networks for each party, where one network captures weakly shared representations and the other learns domain-specific representations, and three loss terms are proposed to enforce the desired learning objectives. Wang et al. [110] presented TrFedDis, a trailblazing network that advanced FL through feature disentangling and uncertainty-aware decision fusion, thereby boosting both efficacy and dependability in non-IID domain features.

iii) IRM: It is an innovative learning paradigm [111] that aims to estimate nonlinear, invariant, and causal predictors from multiple training environments to achieve robust generalization to OOD data. In DL, IRM consists of a feature encoder (g) and a classifier (h). Here, domain $iS^i = \{(x_j^i, y_j^i)\}_{j=1}^{n_i}$ is collected under multiple source domains $i \in I$. The formulation of IRM is to minimize the loss function

$$\min_{h, g} \sum_i \mathcal{R}^i(h \circ g),$$

$$\text{subject to: } h \in \arg \min_{h'} \mathcal{R}^i(h' \circ g), \text{ for all } i \in I \quad (26)$$

where $\mathcal{R}^i(f) := \mathbb{E}_{x^i, y^i} [l(f(x^i), y^i)]$.

Arjovsky et al. [111] treat the classifier h as a fixed scalar, resulting in the optimization problem formulation as follows.

Francis et al. [112] utilized IRM to learn invariant predictors across client domains, achieving optimal empirical risk on all participating clients. The central objective of the FedGen [113] framework was to learn invariant predictors by aggregating local models and masks from participating clients, utilizing a penalty term similar to IRM as a regularizer to discourage overfitting to specific data distributions and to encourage generalization across distributions. Zhang et al. [114] introduced distributionally robust neural networks, DANNs, and MMD feature learning, as well as referencing previous work on invariance methods such as correlation alignment and IRM, providing a comprehensive evaluation of these approaches; de Luca et al. [115]

proposed a federated version of IRM that follows a similar procedure to FedAvg, but differs from other federated IRM approaches (such as CausalFed and CausalFedGSD) that involve data communication either through intermediate representations or a shared subset of data, making Fed-IRM distinct in terms of data sharing.

Overall, IRM could be used in FDG to solve domain shift and ensure model robustness across different domains, which aims to learn representations that are invariant to domain-specific variations while still being informative for the task at hand.

iv) *Kernel-based methods*: They provide a powerful framework to address domain shifts and improve generalization performance for FDG. These methods can capture complex relationships and align distributions between different domains by leveraging kernel functions and operating in high-dimensional feature spaces, leading to more robust and transferable models in FL settings.

Tian et al. [57] developed a novel DG method based on gradient aggregation, leveraging the gradient as a kernel mean embedding to align distributions across domains and improve model generalization capability under privacy constraints, achieving better performance and generalization capability under the privacy-preserving condition. Huang et al. [116] introduced the FL neural tangent kernel, a novel analytical framework for overparameterized ReLU neural networks trained in FL, which theoretically converged at a linear rate to a global optimum and demonstrated promising generalization capabilities. Different distributions of observed data from multiple sources are modeled separately, and a kernel-based method was used to adaptively learn their similarities [117], resulting in an adaptive factor that measures the similarity between the distributions. In the collaborative learning setting among distributed clients facilitated by a central server, Salgia et al. [118] introduced a kernel-based bandit algorithm using surrogate Gaussian process models, achieving near-optimal regret performance, and demonstrated the effectiveness of using sparse approximations to reduce communication overhead among clients. Hong and Chae [119] proposed eM-KOFL and pM-KOFL algorithms for online FL to achieve near-optimal performance while minimizing communication costs, with pM-KOFL demonstrating comparable performance to vM-KOFL (or eM-KOFL) across different online learning tasks.

b) *Feature disentanglement-based FDG*: Feature disentanglement is a widely adopted strategy for mitigating challenges related to domain shift and negative transfer when handling different domains, aiming to separate the domain-invariant features and domain-specific features from training samples [25], [26]. In FDG, feature disentanglement-based methods extend the principles of feature disentanglement to the FL setting and aim to learn disentangled representations by leveraging the distributed data from multiple source domains while ensuring privacy preservation and data locality.

Table 2 Summary of Conceptual and Practical Distinctions Between FDG and FDA, Including Their Objective, Target Domain Data, Application Scenarios, and Common Features

Characteristics	FDG	FDA
Objective	Enhance model generalization to unseen target domains	Adapt source domain knowledge to target domain distributions
Target Domain Data	No access to target domain data during training	Known or partially known during training
Applicable Scenarios	Completely unseen target domains	Scenarios with known or partially known target domains but privacy is limited
Common Features	Address domain shift; operate within the FL framework	Address domain shift; operate within the FL framework

Peng et al. [40] significantly advanced the field by adapting adversarial adaptation techniques to the specific constraints of FL. This adaptation, which utilized feature disentanglement, successfully tackled the dilemmas associated with domain shift while preserving privacy. Wu and Gong [19] optimized domain-invariant feature extractors for central aggregation and domain-specific classifiers for central ensembling, allowing for selective knowledge disentanglement between domain-invariant feature representations and domain-specific classification information.

By disentangling the features and capturing the domain-invariant factors, feature disentanglement-based FDG methods enable the trained model to generalize well to unseen domains by leveraging the shared knowledge across domains. Such strategies are instrumental in overcoming domain shift problems and enhancing model transferability within FL contexts.

Beyond the representation learning-based FDG approaches previously discussed, several other techniques have garnered attention. Huang et al. [120] introduced FCCL+, an innovative FL strategy that mitigates model heterogeneity and catastrophic forgetting through the use of cross correlation and instance similarity alongside nontarget distillation, thereby enhancing discriminability and generalization across varied environments. Song and Liu [121] implemented a pseudo-siamese network within local clients to assess discrepancies between client-specific and global models, improving the delineation of the feature space for fault diagnosis models. This method ensures that the feature representations learned do not excessively conform to the idiosyncrasies of the local domains, thus preserving efficacy in unseen domains.

2) *FDA*: FDA is an approach that synergistically combines FL and DA techniques to mitigate the problem of domain shift in a decentralized setting, thereby improving model performance when source and target domains exhibit dissimilar distributions or come from distinct domains. FDG and FDA differ in their task assumptions, objectives, and methodologies while sharing commonalities in addressing domain shift within FL, as shown in Table 2. FDG focuses on developing models that can generalize robustly to entirely unseen target domains, assuming no access to target domain data during training.

In contrast, FDA aims to transfer knowledge from source domains to target domain distributions by leveraging partial access to target domain data while adhering to privacy constraints. Despite these differences, both FDG and FDA are built on the FL framework, employing methods such as feature alignment and adversarial training to address domain shift. They also share common challenges, including privacy preservation and distributed optimization. Consequently, while their assumptions and methodologies differ, both approaches seek to tackle the challenge of domain shifts in FL, making them complementary within the broader scope of domain-aware FL.

a) Unsupervised domain adaptation: UDA is an ML technique designed to adapt models from a source domain, where labeled data is available, to a target domain that lacks labeled data. In FDG, UDA techniques aim to learn DIRs that capture shared knowledge across domains while reducing domain-specific differences. The goal is to bridge the domain gap and enable models to generalize well to unseen domains without access to labeled data in those domains.

Peng et al. [40] proposed an unsupervised FDA method that aligns learned representations across different domains with the data distribution of target domains to solve domain shift. Wu and Gong [19] proposed a new approach called collaborative optimization and aggregation by employing hybrid batch-instance normalization and collaboration of frozen classifiers to optimize a generalized target model for decentralized DG and UDA. Zhuang et al. [122] introduced FedFR, a method for UDA in face recognition, which combines clustering-based DA and FL to improve performance on a target domain with different data distributions from the source domain. Zhou et al. [123] introduced STU-KD, a scheme for privacy-preserving adaptation of a compact model to edge devices, which combines DA techniques with KD to efficiently transfer knowledge from a large target model to the compact model. The proposed DualAdapt method [94] is an FDA technique that enables the model to adapt to the target domains while preserving knowledge learned from the source domains, achieving high accuracy while minimizing communication costs and computational resources required on clients. Niu et al. [124] introduced a novel federated UDA strategy termed knowledge filter to adapt the central model to the target data when unlabeled target data is available. Qiu et al. [125] proposed the federated semi-supervised learning method, employing federated pseudo-labeling for distributed medical image domains, which enhances FDG by effectively utilizing unlabeled data.

In summary, UDA in FDG is pivotal for mitigating domain shift issues and boosting model transferability across federated domains. This enables models to achieve strong performance in unfamiliar domains, even with limited labeled data.

b) Zero-shot and few-shot adaption: Zero-shot and few-shot adaptation techniques tackle data scarcity and

bolster model generalization. These methods empower models to function effectively in tasks or domains with minimal or no explicit training data, thereby enhancing adaptability to novel scenarios. In FDG, zero-shot and few-shot adaptation techniques aim to enhance model generalization across distributed domains.

Yang et al. [126] introduced zero-shot adaptation with estimated statistics to solve the difficulties of FDG, where the zero-shot adapter facilitates the learned global model in directly bridging a significant domain gap between seen and unseen clients during inference. Huang et al. [127] tackled the issues of few-shot, model-agnostic FL by unveiling a framework that utilizes public datasets to enhance performance in the face of limited private data. This approach effectively navigates issues like inconsistent labels and domain gaps by employing model-agnostic FL techniques and latent embedding adaptation.

Zero-shot and few-shot adaptation techniques in FDG tackle domain shifts and privacy concerns, thereby enhancing the performance of FL in real-world scenarios characterized by limited data access and distribution constraints.

c) Adversarial domain adaptation: Adversarial domain adaptation can also be applied in FDG, where the goal is to learn models that can generalize well across multiple domains without explicitly sharing data. In this scenario, each domain has its data distribution and potentially different labeling conventions or data biases.

Zeng et al. [128] introduced a novel strategy, one-common-source adversarial domain adaptation, aimed at addressing domain shifts in each target domain (private data) by leveraging a common source domain (public data). This strategy includes ADA with gradient matching loss for pretraining encoders. Zhao et al. [39] developed a federated multisource DA method that integrates transfer learning with FL for machinery fault diagnosis, prioritizing data privacy. This method employed federated feature alignment to reduce feature distribution discrepancies and introduced a joint voting mechanism for refining the global model using pseudo-labeled samples. Consequently, this method achieves precise identification of target data while ensuring the protection of data privacy. Peng et al. [40] pioneered an approach to FDA by synchronizing the learned representations among distributed nodes with the target node's data distribution. This method leverages adversarial adaptation techniques, a dynamic attention mechanism, and feature disentanglement, thereby facilitating knowledge transfer and confronting the domain shift task inherent in FL. Zhang et al. [99] proposed FedADG, a federated adversarial DG strategy that harmonizes distributions across various domains through a reference distribution. This method guarantees a universal feature representation that generalizes well, simultaneously safeguarding local data privacy.

By incorporating ADA into FDG, models can learn DIRs that capture common features across multiple domains, enabling the final global model to generalize well to

unseen domains, even in the absence of explicit data sharing.

d) *Gradient consistency DA*: Gradient consistency [129] refers to maintaining alignment in the direction and magnitude of model gradients across diverse data domains during parameter optimization. In FDG, the goal of gradient consistency is to ensure that, despite the distributed and heterogeneous nature of the data, the model can learn a unified representation that generalizes well across all domains through gradient alignment.

Zeng et al. [128] proposed a method called gradient matching FDA, which aims to minimize domain discrepancy by utilizing a public image dataset and training resilient local federated models specifically for target domains. In [128], the gradient matching loss L_{GM} is expressed as the expected cosine distance between $\dot{g}_s(\vartheta)$ and $\dot{g}_t(\vartheta)$, considering all possible values of ϑ

$$L_{GM} = \mathbb{E}_{\vartheta} \left[1 - \frac{\dot{g}_s(\vartheta)^T \dot{g}_t(\vartheta)}{\|\dot{g}_s(\vartheta)\|_2 \times \|\dot{g}_t(\vartheta)\|_2} \right] \quad (27)$$

where $\dot{g}_s(\vartheta)$ represents the expected gradient vector from the source domain, $\dot{g}_t(\vartheta)$ represents the expected gradient vector from the target domain, and the gradients are computed concerning the compatibility model parameters ϑ . The Euclidean norm is represented by $\|\cdot\|$, and the expectation over the compatibility model parameters ϑ is denoted by $\mathbb{E}_{\vartheta}$. Minimizing the gradient matching loss L_{GM} aims to align the gradient vectors between the source and target domains, facilitating DIR learning and improving the generalization capability across federated domains.

Zhu et al. [130] developed an approach to mitigate the issue of domain shift in FL by utilizing gradient covariance and the geometric mean of Hessians. This strategy effectively captures the consistency across and within silos. Compared to alternative DA methods, this approach demonstrates superior performance. The gradient covariance between two random variables X and Y is defined as follows:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] \quad (28)$$

where $E[X]$ is the expected value (mean) of X and $E[Y]$ is the expected value of Y . The expected value operator $E[\cdot]$ calculates the mean of its argument. By leveraging gradient covariance, it is possible to facilitate the integration of learning from multiple tasks or domains by aligning gradient directions, thereby improving model generalization and performance on a wider range of tasks.

Wei and Han [131] introduced a multisource collaborative gradient discrepancy minimization method for FDG. This approach facilitates intradomain and interdomain gradient matching across various source domains, with the objective of developing a model that remains unbiased toward specific domains and exhibits superior performance

on unseen target domains. Specifically, the method employs intradomain gradient matching to synchronize the gradients of classifiers for original and augmented images from isolated source domains. This strategy aims to steer local models toward capturing the essential semantic content present in both types of images

$$L_{gm}^{\text{intra}} = 1 - \text{sim}(g_i, g'_i) \quad (29)$$

where $\text{sim}(\cdot)$ denotes the cosine similarity measure, g_i represents the gradient of classifier C_i with respect to original images, and g'_i corresponds to the gradient of classifier C_i for augmented images.

To mitigate the domain gap across decentralized source domains, interdomain gradient matching leverages classifier heads from other domains as intermediaries. This method aims to minimize gradient discrepancies between the classifier head in use and those from disparate domains, thus reducing domain shift and bolstering model generalization across federated domains

$$L_{gm}^{\text{inter}} = \sum_{j=1}^n (1 - \text{sim}(g'_i, g_j^{t-1})) \quad (30)$$

where g'_i denotes the gradient of classifier C_i with respect to augmented images and g_j^{t-1} represents the gradient of classifier C_j^{t-1} from the j th source domain concerning original images. In addition, this approach can be adapted for FDA tasks through the refinement of the target model using pseudo-labeled data from the target domain.

These studies demonstrate innovation and diversity in addressing domain variability in distributed data domains by adopting gradient consistency as a core mechanism. Although each method has its unique implementation and focus, they collectively highlight the importance of achieving domain-invariant learning and generalization capabilities in FDG.

3) *Federated Transfer Learning*: It is a valuable approach for scenarios [21] where two datasets differ not only in their samples but also in their feature spaces. It allows models to leverage knowledge learned from one dataset to improve the performance on a different dataset with distinct features while preserving data privacy and security. FTL can also be applied to FDG, where the goal is to improve model generalization across multiple federated datasets that exhibit domain shifts and feature variability. The objective of applying FTL in the context of FDG is as follows:

$$\Theta^* = \arg \min_{\Theta} \sum_{i=1}^n \delta_i * \mathcal{L}(\Theta, D_i) + \beta * F(\Theta) + \sigma * \Omega(\Theta) \quad (31)$$

where Θ denotes the model parameters to be optimized, Θ^* represents the optimal model parameters, n signifies the number of clients within the FL system, $\mathcal{L}(\Theta, D_i)$ is the loss function that measures the discrepancy between the predicted outputs of the model and the ground truth labels on client i 's data D_i , δ_i is the weight assigned to the loss function $\mathcal{L}(\Theta, D_i)$ for client i , β is a regularization hyperparameter, $F(\Theta)$ is the regularization term that encourages model parameter smoothness or complexity control, σ is a hyperparameter that controls the tradeoff between the regularization term and the DG term, and $\Omega(\Theta)$ is the DG term that aims to minimize the distribution discrepancy between different clients' data distributions promoting generalization across domains.

Chen et al. [56] developed the D-WFA framework, which employs discrepancy-based weighted federated averaging to collaboratively train a global diagnosis model. This approach leverages locally labeled source domain data and unlabeled target domain data, prioritizing data privacy. Zhang and Li [60] introduced an FTL method for fault diagnosis that addresses data privacy and the domain shift phenomenon by utilizing unique models for different users. This method features a federated initialization stage and a federated communication stage powered by deep adversarial learning. Zhang and Jiang [76] proposed another FTL method for machinery fault diagnostics. This method uses DL-based models trained locally at each client to ensure data privacy and introduces a novel DA approach to bridge domain gaps without sharing local data, showing efficacy in real industrial settings. Li et al. [132] offered the FedSWP framework, enhancing the smart work packaging system for construction occupational health and safety management. This framework focuses on protecting construction workers' personal image information, providing accurate and personalized safety alerts and healthcare solutions while addressing privacy concerns. Zhou et al. [133] introduced a bearing faulty prediction method that combines FTL with KD. This method, which includes signal-to-image conversion, multisource FTL, and multiteacher-based KD, significantly improves generalization capability and achieves higher accuracy in bearing fault prediction tasks with a lower parameter size. Fan et al. [134] presented IoTDefender, a personalized and distributed intrusion detection framework for 5G IoT. Leveraging FTL and 5G edge computing, this framework tackles the issues of heterogeneity, data isolation, and limited data availability. It achieves high detection accuracy and generalization capability with lower false positive rates, all while preserving user privacy. Chen et al. [135] showcased FedHealth, an FTL framework for wearable healthcare. This framework tackles data isolation and the demand for personalization by aggregating data via FL and developing personalized models. It showcases accurate and privacy-preserving healthcare applications, including wearable activity recognition and Parkinson's disease diagnosis.

By leveraging transfer learning techniques within an FL, models can learn from multiple domains while preserving data privacy, enabling improved generalization capabilities and robustness in diverse federated environments.

4) *Channel Decoupling*: Channel decoupling is a technique used in ML, particularly in DG, to disentangle domain-specific information from shared information in feature representations. It can be used in FDG to disentangle domain-specific knowledge from shared knowledge by decoupling channel-wise statistics, enabling better generalization across unseen domains by reducing domain-specific variations.

Yang et al. [126] proposed a method for client-agnostic learning with mixed instance-global statistics. In this approach, local models aim to learn client-invariant representations by optimizing client-agnostic objectives using augmented data. Feng et al. [136] put forward the FedMRI algorithm, which partitions the model into a shared encoder and client-specific decoders. This architecture preserves domain-specific characteristics while fostering a generalized representation, effectively tackling domain shift issues. Shen et al. [137] introduced the CD2-pFed framework, employing cyclic distillation-guided channel decoupling to tailor the global model for individual clients within FL. This approach effectively navigates the complexities of heterogeneous data distributions, enhancing generalization across clients. Tan et al. [49] developed FedStar, a federated graph learning framework that tackles the non-IID problem in graph data. By leveraging structure embeddings and independent structure encoders, FedStar extracts and shares essential structure information, securing structure-based domain-invariant knowledge and mitigating feature misalignment. Weng et al. [138] proposed FedUCC, an innovative federated unsupervised cluster-contrastive learning method for Person ReID. FedUCC employs a three-stage strategy to discover generic, specialized, and patch knowledge, ensuring cross-domain consistency and maintaining local domain-specific insights.

By allocating distinct channels to each domain, the global model is capable of learning domain-specific representations, thereby efficiently overcoming the issues associated with heterogeneous data distributions among the participating domains. Channel decoupling allows for fine-grained adaptation and enhances the model's ability to generalize across diverse domains in FL.

5) *Style Transfer*: Style transfer in FDG refers to the process of transferring the style or visual characteristics of a source domain to a target domain in the FL setting. The goal is to generalize the learned style representations across domains while preserving the content of the target domain.

Nergiz [139] introduced a federated adaptation of the neural style transfer algorithm as a data augmentation technique to solve the issues of data collaboration in computational pathology, specifically focusing on the highly class-imbalanced Chaoyang colorectal cancer

imaging dataset while ensuring privacy preservation without any data leakage. Chen et al. [69] introduced a DG method called CCST for image recognition in FL, which surpasses state-of-the-art methods on DG while being compatible with existing DG techniques. Shenaj et al. [68] introduced FFREEDA, a task for semantic segmentation in autonomous driving, proposing the LADD algorithm that utilizes pretrained model knowledge through self-supervision and a federated clustered aggregation scheme based on clients' style, demonstrating superior performance compared to existing approaches. Fantauzzo et al. [140] presented FedDrive, a benchmark for FL in semantic segmentation for autonomous driving, addressing difficulties of statistical heterogeneity and DG, evaluating state-of-the-art algorithms with style transfer methods to enhance generalization capabilities.

It is important to note that style transfer for FDG is an evolving research area, and the specific techniques and methodologies may vary depending on the latest advancements.

6) *Federated Meta-Learning*: Federated meta-learning combines the concepts of FL with meta-learning to generate a global model across various domains. This approach enables the model to swiftly adapt to new domains with minimal data, promoting robust generalization in a federated setting.

a) *Model-agnostic meta-learning*: MAML [141] is a meta-learning approach that aims to learn an initial model that can quickly adapt to new tasks with limited labeled data. In FDG, MAML is utilized to overcome the obstacle of generalization across diverse federated domains with varying data distributions. MAML targets domain shifts by fostering a meta-level optimization process that facilitates rapid adaptation of models to new tasks using limited data. Instead of training a model from scratch for each new task, MAML algorithms aim to learn a good initialization or set of initial parameters that can be fine-tuned or adapted to new tasks with minimal data. The specific form of the objective function may vary depending on the MAML algorithm used, but in general, it involves a two-level optimization process: inner loop optimization and meta-optimization. In the inner loop optimization, for each task or episode, the model is trained on a support set of labeled examples specific to that task. The goal is to update the model parameters to minimize task-specific loss or error on the support set. This is typically done through gradient descent or other optimization techniques. In the meta-optimization, the meta-parameters of the model are updated based on the performance of the model on the query sets of multiple tasks. The objective is to find meta-parameters that can generalize well across tasks and enable fast adaptation. The meta-parameters are updated using the gradients computed from the meta-objective, which is often defined as the average loss or error across the query sets of multiple tasks.

1) *Inner Loop Optimization*: The base learner is fine-tuned using support samples D_{S_i} from the learnable initialization Φ for a fixed number of weight updates via gradient descent. The task adaptation objective is to minimize the loss function $L(D_{S_i}, \Phi_{i,j})$ using gradient descent. At the j th step of inner loop optimization, the base learner parameters are updated as

$$\Phi_{i,j+1} = \Phi_{i,j} - \alpha \nabla_{\Phi_{i,j}} L(D_{S_i}, \Phi_{i,j}) \quad (32)$$

where α is the learning rate and $\Phi_{i,j}$ represents the learner's parameters at the i th task and the j th step of the inner loop.

2) *Inner Loop Optimization Iterations*: After performing J number of inner loop update steps, the task-specific base learner parameters become $\Phi_{i,J}$.

3) *Meta-Optimization*: The meta-learned initialization Φ is evaluated based on the generalization performance of the task-specific base learner with parameters $\Phi_{i,J}$ on unseen query examples D_Q . The objective of MAML is to minimize the meta-learning algorithm's objective function, denoted as $L(D_Q, \Phi_{i,J})$, as follows:

$$\Phi = \Phi - \zeta \nabla_{\Phi} L(D_Q, \Phi_{i,J}) \quad (33)$$

where ζ is the meta-learning rate.

Chen et al. [142] presented FedMeta, a federated meta-learning framework that tackles both statistical and systematic complications in FL. By reducing communication costs, enhancing convergence speed, increasing accuracy, and ensuring privacy, FedMeta surpasses existing optimization algorithms. Wang et al. [143], inspired by MAML, proposed GraphFL methods to navigate the non-IID issue in graph data and new label domains, employing a structure that divides each task into a labeled training set and a separate query set. Jiang et al. [144] introduced a personalized FL approach utilizing MAML algorithms. This work elucidates the meta-learning interpretation of the widely adopted FedAvg algorithm and proposes a refined version tailored to enhance FL personalization, ultimately aiming to improve the global model's generalization capability. Lin et al. [54] developed a collaborative learning framework aimed at real-time edge intelligence, leveraging federated meta-learning to train models across multiple source edge nodes and swiftly adapt them to new tasks at target nodes with limited data. Fallah et al. [145] delved into the generalization properties of MAML algorithms within supervised learning. They presented a novel stability definition that accounts for the number of tasks and samples per task, facilitating an in-depth analysis of MAML's generalization error.

Li et al. [146] introduced a cutting-edge pancreas segmentation model that leverages a meta-learning strategy and latent-space feature flow generation. This approach

effectively reduces interference from background clutter and appearance-style discrepancies through a coarse-to-fine workflow, achieving superior performance across three pancreas datasets and outperforming current state-of-the-art generalization methods. Chen et al. [147] introduced FedMeta-FFD, a pioneering method that combines FL and meta-learning to address the few-shot fault diagnosis task in IIoT. This method allows clients to utilize external datasets during the training of a global meta-learner, leading to enhanced convergence speed and accuracy compared to existing techniques. Khodak et al. [148] proposed a theoretical framework that merges task similarity, online convex optimization, and sequential prediction algorithms. This framework enhances few-shot learning and FL performance through adaptive task-similarity learning, improved transfer-risk bounds, and average-case regret bounds in dynamic or varied tasks.

By utilizing MAML, the model parameters can be updated on meta-training data, which consists of raw input data from different federated domains, and then evaluated on held-out meta-test data generated from frequency space with different distributions.

b) *Episodic meta-learning*: Beyond the MAML methodologies, additional approaches are explored in federated meta-learning for FDG. One such method is episodic learning, which can be applied in FDG to improve the transferability of models across different domains. Liu et al. [18], [149] introduced episodic meta-learning for training models in FL across varying data distributions by explicitly simulating domain shifts and employing a meta-learning approach to update model parameters.

c) *Meta critic*: Additionally, the meta-critic network provides feedback on the primary models' performance across different domains, facilitating guided adaptation and enhancing their generalization capabilities. Chen et al. [52] proposed a personalized *federated optimization framework with meta critic* (FedMC) to tackle heterogeneity in federated networks. This framework effectively identifies robust and generalizable domain-invariant knowledge across clients, tackling the issues arising from diverse client data distributions.

7) *Data-Free Methods*: Data-free [150] methods aim to perform learning tasks without using any labeled data from the target domains by leveraging auxiliary information, prior knowledge, or transfer learning techniques to generalize to unseen domains.

a) *Zero-shot and few-shot learning*: Zero-shot learning and few-shot learning aim to recognize or classify objects from unseen domains for which no labeled samples are available during training.

Sun et al. [74] proposed federated zero-shot learning to transfer mid-level semantic knowledge and optimize a generalizable FL model. Sariyildiz and Cinbis [83] proposed a generative model combined with gradient matching loss, transforming zero-shot learning issues into a supervised classification task. This approach significantly boosts

generalized zero-shot classification accuracy, surpassing the performance of prior methodologies. Micaelli and Storkey [100] demonstrated the possibility of achieving zero-shot knowledge transfer through an adversarial approach, where the adversarial generator is trained to produce rigorous images for a student network to match the predictions of a teacher network. Zhang et al. [101] introduced FedZKT, a data-free FL framework that enables devices to autonomously select on-device models based on their local resources, facilitating knowledge transfer among these heterogeneous on-device models through a zero-shot distillation technique. Li and Zhao [151] developed a federated zero-shot fault diagnosis framework that utilizes a semantic knowledge base, a bidirectional alignment network, and cloud-edge collaboration to effectively diagnose both local and global unseen fault categories. Yuan et al. [16] proposed the collaborative semantic aggregation and calibration framework, a pioneering approach that facilitates data-free DG by merging model-based semantic information across multiple domains, ensuring adherence to privacy constraints. Additionally, the framework tackles the issue of semantic dislocation by implementing cross-layer semantic calibration, employing an attention mechanism to enhance the precision of semantic alignment.

By incorporating zero-shot learning and few-shot learning techniques into FDG, models can improve their ability to generalize across multiple domains with limited labeled data.

b) *Knowledge distillation*: Data-free KD [100], [152] is a technique used to transfer knowledge from a large, well-trained model to a smaller, more compact model without the need for labeled training data.

Frikha et al. [150] presented DEKAN, an approach named domain entanglement via knowledge amalgamation from domain-specific networks, which effectively extracts and combines domain-specific knowledge from existing teacher models to create a domain-shift-robust student model. Niu et al. [124] introduced a mutually collaborative KD, enhancing both federated UDA and DG by learning domain-invariant features in a data-free manner through collaborative distillation across disparate local models. Zhu et al. [153] proposed a data-free KD method for heterogeneous FL, which utilizes a lightweight generator to aggregate user information and improve the generalization performance of the global model with fewer communication rounds. Jeong et al. [154] developed data-free federated distillation, a distributed model training algorithm for on-device ML that minimizes interdevice communication overhead. This method tackles the non-IID data problem by implementing federated augmentation, which synthesizes an IID dataset to enhance performance. Zhang et al. [155] introduced FedFTG, a data-free KD technique for fine-tuning the global model on the server, effectively mitigating data heterogeneity in FL and serving as a complementary enhancement to existing local optimizers.

Data-free KD can be leveraged successfully in FDG, wherein the student model is trained using multiple teachers who have been trained on diverse source domains, facilitating the generalization capability to previously unseen target domains.

c) Semantic aggregation and calibration: In FDG, semantic aggregation and calibration refer to techniques used to consolidate and refine semantic representations across multiple nodes in a privacy-preserving manner.

Yuan et al. [16] introduced CSAC, a novel privacy-preserving method that enables separated DG by unifying multisource semantic learning and alignment through an iterative process of data-free semantic aggregation and cross-layer semantic calibration. Niu et al. [124] proposed a data-free semantic collaborative distillation approach aimed at acquiring DIRs, applicable to both federated UDA and DG.

By employing semantic aggregation and calibration techniques, FDG methods can effectively leverage shared knowledge across domains while adapting to the specific characteristics of individual domains, leading to improved generalization performance and robustness in FL scenarios.

8) Federated Domain Translation: Domain translation [156], [157] refers to the task of converting or adapting content from one domain to another domain while preserving its meaning and quality. Federated domain translation is a method that aims to perform translation tasks across multiple domains in an FL framework while preserving data privacy and security.

Iterative naive barycenter [158] for FDG aims to find a common representation or domain-invariant subspace by iteratively averaging the distributions or representations from multiple source domains. Zhou et al. [159] proposed a federated domain translation method that generates pseudo-data for each client, which could be useful for multiple downstream learning tasks. Xie et al. [160] introduced FedMed-GAN, a benchmark for federated domain translation. This benchmark effectively combats mode collapse, preserves generator performance, and showcases adaptability across varying data proportions and distributions.

By translating data or knowledge across domains in the FL setup, models can benefit from a more diverse and comprehensive training dataset, leading to improved performance and generalization in real-world applications.

9) Regularization Strategies: Regularization strategies in FDG aim to improve the generalization performance of models across multiple domains by mitigating the negative effects of domain shift and improving DIRs.

To combat overconfidence and dropping learning curves in pseudo-labeling for FDG, Shenaj et al. [68] proposed using KD loss based on soft predictions and stochastic weight averaging applied to clients' teachers to stabilize the learning curve and prevent the model from forgetting pretraining knowledge. Zhuang et al. [122] presented a

novel domain constraint loss in FL to regularize source domain training where domain constraint loss serves as a regularization term during the training of the source domain model, promoting alignment toward the target domain within the aggregated global model. Zhu et al. [130] proposed a new regularization methodology called federated invariant learning consistency to promote both intersilo and intrasilo consistencies in federated networks by leveraging the domain-level gradient covariance and the geometric averaging of Hessians. In addition, they introduced a novel weighted geometric mean mechanism to calculate the geometric mean with inconsistent signs on gradients. Atwany and Yaqub [66] employed flatness in the training of separated DG, incorporating iteration-wise weight averaging and domain-level gradient variance regularization as effective techniques. Feng et al. [136] developed a weighted contrastive regularization method for the globally shared part of the model, which helps alleviate domain shifts among clients during training and improves convergence. Liu et al. [18] employed the InfoNCE objective to impose regularization, using positive and negative feature pairs, to enhance the domain-invariance and discriminability of the features in their FL model, ultimately improving its performance. Huang et al. [62] implemented consistency regularization to align local instances with their respective unbiased prototypes, markedly enhancing both the effectiveness and efficiency of mitigating domain shifts in FL.

Overall, regularization strategies play a crucial role in FDG by mitigating overfitting and improving the generalization performance of models across different unseen domains.

D. Aggregation Optimization

AO in FDG refines the integration of local models from diverse domains into a globally coherent model, using strategies like weight optimization, gradient optimization, and aggregation policy to enhance performance and generalizability across various domains.

a) Weight optimization: The FedAvg algorithm in FL aggregates global model parameters based on the number of local model parameters contributed by each client, disregarding the quality and importance of the local model parameters, which can adversely impact the performance of the global model. In FDG, optimizing aggregation weights is crucial to overcoming the dilemmas posed by heterogeneous data distributions and nonuniform network architectures across participating clients.

Chen et al. [56] proposed D-WFA, a dynamic weighting strategy based on MMD, adjusting the weights of clients and updating multiple local models with generalization capability. Zhang et al. [161] presented the generalization adjustment method in FL that dynamically calibrates aggregation weights to optimize the objective, achieving a tighter generalization bound by explicitly reweighting aggregation instead of relying on implicit multidomain

data sharing as in conventional DG settings. Yuan et al. [16] introduced a semantic similarity-based attention mechanism that dynamically assigns weights to cross-layer pairs, prioritizing pairs with higher semantic similarity for precise alignment while reducing the emphasis on less similar pairs to prevent semantic dislocation. Yang et al. [126] proposed to dynamically generate instance-wise interpolation parameters for mixing instance and global statistics with a learning-based network to build the global model deployed to unseen clients. Alekseenko et al. [162] proposed a methodology that employs distance measurements to address non-IID data in FL, enhancing model generalization by downgrading the influence of the most divergent client.

Optimizing aggregation weights in FDG enables effective integration of client contributions, accounting for their importance and relevance, resulting in a more adaptive and robust global model capable of performing well across diverse data distributions and network architectures.

b) Gradient optimization: Gradient optimization in FL refers to the process of updating model parameters using gradients computed on local participant data. In FDG, gradient optimization plays a critical role by enabling efficient parameter updates across decentralized domains. It ensures convergence to an optimal global model while accounting for variations in data distributions and network architectures, thereby enhancing the model's generalization and performance across diverse domains.

Federated voting aimed to fine-tune the global model in FL without using the ground labels of the target domain samples, thus improving the effectiveness of feature distribution matching [108]. In the global mutual optimization [163], each domain underwent model distillation to learn from its peers, effectively mitigating any potential negative impacts of domain-specific gradient updates on the overall global model. To enhance the robustness of the global feature encoder across clients, a domain critic [52] was introduced to regularize the feature representations learned by both the local and global feature encoders, utilizing the Wasserstein critic as the chosen distance metric due to its gradient superiority. Tian et al. [57] introduced an innovative gradient alignment loss to enhance the gradient aggregation process on centralized servers, aiming to bolster the model's generalization to novel yet related domains. Zeng et al. [128] developed the gradient matching federated fine-tuning method, which updates pretrained local federated models using the one-common-source adversarial domain adaptation strategy, minimizing the gradient matching loss between sites to drive the optimization direction toward specific local minima. Wu and Gong [19] introduced a prediction agreement mechanism to overcome local disparities toward central model aggregation for achieving better decentralized DG. Zhang et al. [164] introduced a novel gradient correction technique, termed GRACE, leveraging feature alignment regularization within a meta-learning framework on the client side to mitigate the overfitting of personalized gradients.

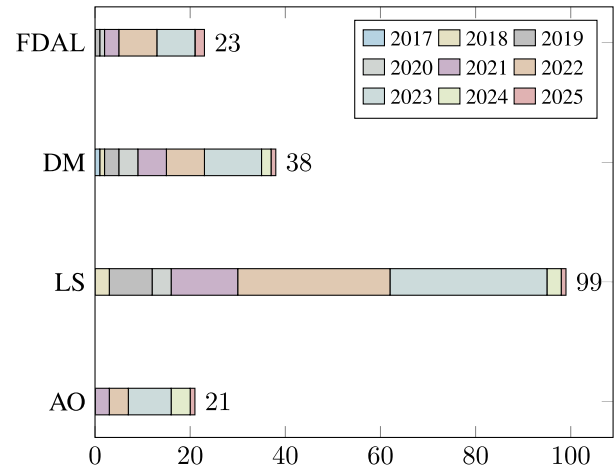


Fig. 7. Publication trends across methods (as of February 2025). The stacked bar chart illustrates the cumulative and annual distribution of publications across four key methodological areas: AO, LS, DM, and FDAL.

In summary, optimizing aggregated gradients in FDG is essential to mitigate domain shift, accommodate data heterogeneity, and enhance model performance across various domains.

c) Aggregation policy: The aggregation policy in FL refers to the strategy or algorithm used to combine the model updates from multiple participating clients and generate an updated global model. In FDG, designing an effective aggregation policy is essential for achieving accurate and robust models.

Shenaj et al. [68] introduced a clustered and layer-aware aggregation policy where global parameters were aggregated over selected clients and cluster-specific parameters were averaged within clusters, where clients were clustered based on their transferred styles using H-means clustering. Cong et al. [165] proposed the FDG-GRMA framework, which leverages a robust model aggregation strategy to construct a global model that both preserves data privacy and efficiently diagnoses faults through strategic parameter and feature transmissions between the central server and source domain clients. Zhang et al. [164] employed a consistency-enhanced reweighting aggregation on the server side to recalibrate the aggregated gradients, thereby improving the global model's generalization capabilities across heterogeneous datasets. Chung and Al Kontar [166] advanced FDG by introducing a reparameterization technique that redefines local entropy, enabling the aggregation of perturbed local gradients to approximate global gradients, thus facilitating robust and generalized model learning across clients without data sharing. Song and Liu [121] proposed an aggregation strategy on the central server that involves evaluating each local model's performance across other source clients, assigning higher weights to models with poorer generalization to ensure that the final aggregated model exhibits enhanced performance across all domains.

Liu et al. [167] introduced a dynamic aggregation strategy that adaptively weights clients based on domain-invariance principles, which helps to mitigate domain variations and enhance the DG capability of the server model. Chen et al. [168] introduced FedAWA, a novel aggregation method that replaces traditional data-driven weighted averages with arithmetic averaging. The approach critiques existing aggregation weight strategies and integrates confidence- and fairness-based adjustment mechanisms to enhance model reliability and ensure balanced client contributions and empirically demonstrates the method's effectiveness, thus advancing the field of FDG.

Zhang et al. [169] proposed FedCD, a local training strategy leveraging cross-client knowledge fusion and a global weighted aggregation mechanism to enhance generalization and robustness, with theoretical analysis and extensive experiments showing its superior performance over state-of-the-art FL models. However, increased client numbers may lead to significant computational and communication overheads that could offset the benefits of personalization.

Optimizing the aggregation policy effectively helps to combine and utilize knowledge from diverse domains, ensuring the generation of a robust and adaptable global model capable of performing well across different domains and scenarios.

E. Dissecting Research Dynamics

Fig. 7 depicts the publication trends across various methodologies, identified as AO, LS, DM, and FDAL, from 2017 to 2025. Fig. 7 employs a stacked bar chart to elucidate the cumulative publication output per methodology alongside the yearly distribution. A comparative analysis of the bars reveals a marked increase in publications across all subareas, indicative of a growing interest in FDG research. The color-coded stratification within the bars enables an expedient evaluation of the relative volume of research among the subareas, highlighting shifts in the focus of the research community. It is particularly noteworthy that the LS methodology has commanded a significantly higher interest in research, as demonstrated by its dominant proportion of the total publications.

IV. APPLICATIONS

FDG has broad applications in scenarios where data are distributed across multiple domains, and data sharing is limited due to privacy concerns or regulatory reasons. This section expands on some of the popular tasks and applications for FDG, highlighting specific challenges, real-world outcomes, and FDG's performance in each context.

A. Healthcare

In healthcare, each hospital collects medical data from its patients, which can vary significantly in terms of demographics, medical conditions, and treatment protocols. However, the common goal is to develop a model that

can predict disease outcomes across hospitals with varying data distributions. FDG addresses this challenge by enabling local training at each hospital, preserving data privacy [44], [170], [171]. For instance, Luo et al. [172] demonstrated that FDG can enhance the generalization of COVID-19-like pandemic prediction models across hospitals by employing generative DA methods. In the multi-hospital case study, FDG improved pandemic prediction accuracy compared to other learning approaches while ensuring compliance with privacy regulations. However, a major bottleneck in healthcare applications is the limited availability of labeled data, which can significantly hinder the performance of FDG. To mitigate this challenge, researchers are exploring techniques such as semi-supervised learning and data augmentation [172], which leverage both labeled and unlabeled data to improve model generalization.

B. Finance

The finance sector involves highly sensitive data that are distributed across various institutions, such as banks, insurance companies, and payment processors [173], [174], [175]. Given the inherent privacy concerns, FDG offers significant advantages in training models on these distributed datasets while ensuring confidentiality and enhancing model generalization. Financial models trained on data from a single institution may face challenges in generalizing to other institutions due to differences in data distribution, business processes, and regulatory requirements. FDG facilitates cross-domain collaboration by training models on local data and aggregating domain-invariant knowledge. For example, the work [176] applied FDG to financial risk analysis by federated meta-learning, effectively learning discriminative feature extraction on unseen class samples, outperforming existing meta-based approaches in the fraudulent credit card detection task. However, addressing data heterogeneity [176] and generalization to novel categories [175] is vital for advancing FL in finance. Future research should prioritize improving model adaptability, reducing heterogeneity effects, and developing innovative algorithms and data techniques to enhance fraud detection. These improvements will strengthen the real-world application of FL in the financial sector.

C. Education

Educational institutions are increasingly engaging in international collaborations, enabling students to connect and learn from peers in different countries, thereby promoting cultural understanding and global perspectives. However, these institutions must adhere to privacy and data protection regulations when collecting, storing, and analyzing student data. FDG can facilitate international collaboration among educational institutions while ensuring data privacy, such as in grade classification [177] or

digital educational environments [178]. For instance, models trained on data from one institution often face challenges when applied to other institutions due to differences in curricula, teaching methods, and student demographics. FDG can help develop models that generalize across educational domains by learning domain-invariant features. As proof, the research [177] proposed an FTL framework for student performance classification, addressing both data privacy concerns and the need for collaboration among institutions without sharing sensitive data. Additionally, by integrating heterogeneous DA techniques, the framework enables effective knowledge transfer between nodes with different data distributions and scales. This is especially beneficial in educational settings where data characteristics may vary significantly. However, challenges such as small sample sizes per institution and data fragmentation persist. Research on adaptive models for data sparsity suggests that FDG can mitigate these issues to some extent, although further exploration is needed in highly diverse data environments.

D. Transportation

The transportation sector plays a vital role in facilitating the movement of people and goods within and between cities, regions, and countries. Transportation data are typically collected and managed by transportation authorities, government agencies, transportation service providers, and technology companies, which is crucial to ensure the security and privacy of information. FDG in the transportation domain can facilitate data collaboration between cities and regions; support the design and optimization of intelligent transportation systems [179], [180], autonomous drive [68], and optimal transport [181]; and enhance capabilities for traffic safety [179]. Namely, models trained on data from a certain city are often difficult to promote in other cities due to differences in traffic patterns, road networks, and vehicle types. FDG helps develop models that can generalize across multiple cities and regions by learning domain-invariant features. For example, Wang et al. [179] proposed that the hierarchical FL approach allows for the development of models that can be generalized across different geographic regions. This generalization means that cities do not need to develop unique models for each region to implement the system, simplifying the deployment process and making it more adaptable to a variety of urban environments. However, a major bottleneck in the field of transportation is the high degree of variation in data quality and availability across regions, which may limit the effectiveness of FDG. Future research could focus on developing robust data fusion and augmentation techniques to address the variability in data quality and availability across regions, thereby enhancing the effectiveness of FDG in transportation systems.

E. Natural Language Processing

NLP tasks typically rely on vast amounts of textual data, which can originate from highly diverse sources,

encompassing various languages, dialects, cultural contexts, and application domains (e.g., law, medicine, and social media). These variations can lead to significant shifts in the distribution of the training data, making it challenging for traditional models to perform effectively in unseen domains. Furthermore, NLP tasks across different domains often have distinct requirements and objectives, increasing the complexity of cross-domain training. FDG addresses these challenges by enabling knowledge sharing across domains, effectively mitigating issues such as data heterogeneity, cross-domain training complexity, and data privacy protection in NLP tasks [182], [183]. However, it still faces challenges in privacy protection, technical maturity, auditing, non-IID data handling, and result generalizability. In the future, it needs to narrow the performance gap, enhance system efficiency, improve privacy measures, develop personalized approaches, optimize user selection, and establish a comprehensive evaluation framework to facilitate the practical application of FDG in NLP [184].

F. Robotics

Robotics technology faces several challenges, including variations in data distribution due to environmental differences, data privacy and security concerns, data silos, uneven data distribution, and limited computational resources. These factors hinder traditional models' ability to maintain robustness in dynamic environments and effectively leverage distributed data. FDG offers an effective solution by addressing issues such as environmental diversity, data privacy protection, data fragmentation, and resource constraints. FDG has paid attention to the robotic areas, such as learning autonomously [185] to smart robotic wheelchairs [186], learning basic table tennis movements from human inputs [187], cloud robotic systems [188], and object recognition and grasp planning in surface decluttering [189]. As an example, Casado and Demiris [186] proposed an approach that integrates FL with Learning From Demonstration, applied to the smart wheelchair system. It successfully achieves privacy protection, cross-domain generalization, real-time learning, and personalized adaptation and validates its effectiveness in multiuser collaborative training and task completion, advancing the field of assistive robotics. FDG enhances the generalization capabilities of robots in unseen domains while safeguarding data privacy and accommodating distributed computing requirements. However, it still faces challenges, including data heterogeneity, complex cross-domain training, resource limitations, and a shortage of labeled data, necessitating further research to optimize these areas and advance its application in robotics.

G. Industrial Edge Intelligence

Intelligent instrumentation and measurement technology has advanced rapidly, which has resulted in a significant increase in industrial data. This has led to the inevitable arrival of the "era of industrial big data" [190],

[191]. However, industrial data often contain sensitive information (such as production processes and equipment status), and the industrial environment is complex and dynamic. The data distributions across different factories, production lines, or devices may vary significantly, leading to poor model performance in cross-domain scenarios. FDG provides an effective solution for industrial edge intelligence, addressing challenges such as environmental diversity, data privacy protection, data silos, and resource constraints. To illustrate, Chen et al. [56] proposed a local training method for fault diagnosis in FL, incorporating an MMD-based strategy to address domain shifts. Additionally, they introduced a dynamic, discrepancy-based weighted federated averaging algorithm designed to adjust the contributions of client models, accounting for potential domain shifts among clients. The proposed method outperforms other FTL techniques, ensuring data privacy while achieving performance that closely matches centralized learning outcomes, making it highly suitable for industrial applications. Industrial edge intelligence encounters substantial challenges, including data scarcity, non-IID data, and stringent real-time requirements [60], [147]. Future research should prioritize advancements in few-shot learning, non-IID data processing, communication efficiency, multimodal data fusion, and adaptive model updates to enhance the robustness, efficiency, and applicability of industrial edge intelligence systems.

H. Distributed Computing Continuum Systems

The advent of edge computing, complementing cloud computing, has catalyzed the creation of distributed computing continuum systems [192], exploiting the cloud's expansive resources along with the edge's diversity and immediacy. However, current research on edge computing and the distributed computing continuum primarily targets niche problems, yielding solutions of restricted scope. Traditional computer system architectures fall short of encapsulating the complexity of distributed computing continuum systems, characterized by their heterogeneous devices and networks. Since the functional requirements of these systems may evolve, services may be dynamically added or changed, or unexpected events may occur, resulting in alterations to the underlying infrastructure configuration. This necessitates a new and innovative representation of distributed computing continuum systems that is not limited by outdated architectural models. Furthermore, several research works emphasize the need to distribute functionalities and processes among computing continuum entities in a decentralized manner [193]. Therefore, this has opened the possibility of developing FDG in distributed computing continuum systems for maintaining data privacy and security, enabling cross-domain model generalization, and enhancing system performance and robustness in distributed computing environments.

V. FUTURE RESEARCH DIRECTIONS

This article offers an extensive and meticulous survey of the existing literature on FDG, highlighting its inherent advantages as a fusion of FL and DG, resulting in privacy-preserving, scalable, and diverse models capable of effective generalization across multiple domains. To be precise, FDG is a promising technological breakthrough as a subfield within ML that aims to develop models capable of generalizing well across multiple unseen domains while achieving privacy preserving. Most technological developments come with benefits, challenges, and limitations, and FDG is not exceptional. In the near future, several challenges need to be overcome in order to retain more benefits from FDG. One of the primary difficulties faced in FDG is domain shift, which occurs when a model trained on data from one domain performs poorly on data from a different domain due to significant differences in statistical properties. Additionally, data heterogeneity, limited data availability, high communication overhead, and privacy concerns pose significant dilemmas for FL. To effectively tackle these issues, a comprehensive identification and analysis of these challenges is crucial, accompanied by the pursuit of novel theoretical and technical solutions. Within this context, this study examines key problems in FDG, including privacy-preserving FDG, communication-efficient FDG, computation-efficient FDG, heterogeneity, label shift, scalable FDG, continuous FDG, and FDG to novel categories.

A. Advanced FDG Techniques

In the rapidly evolving field of FDG, the development of advanced technologies remains paramount for enhancing the robustness and adaptability of models across various unseen domains. Current research in this area is relatively sparse and insufficient to address the complexities of the issue thoroughly. For instance, tradeoffs such as sacrificing security for DG are prevalent challenges [18], [19]. Additionally, the incorporation of incremental and lifelong learning to adapt to unseen data and tasks promises enhanced generalization capabilities [194]. Moreover, the interplay between personalization and generalization requires careful consideration to determine the optimal model for specific scenarios [164], [195]. Also, ensuring robustness and fairness is crucial to maintaining satisfactory performance in the face of challenges while preventing biases or unfair decision-making processes in models [196]. Furthermore, integrating model interpretability or explainability within FDG can lead to greater transparency, trustworthiness, and ease of debugging, addressing the common issue of opacity in ML decisions [197]. Exploring these advanced technologies not only expands the viable boundaries of DG but also enhances the practical applicability of FDG in real-world scenarios where domain shifts are prevalent.

B. Privacy-Preserving FDG

While FL inherently ensures privacy by retaining data within local domains, FDG faces unique privacy protection challenges that differ from traditional FL. Among these challenges, one significant issue is the variation in cross-domain privacy requirements, as FDG spans multiple domains that may exhibit differences in data sensitivity and privacy regulations [198]. Additionally, FDG involves a fundamental tradeoff between generalization and privacy. While models need to perform well in unseen domains, existing privacy-preserving mechanisms, such as differential privacy, often compromise this ability, making it an ongoing challenge to balance privacy protection and generalization performance [16], [18]. Furthermore, existing aggregation mechanisms are difficult to cope with interdomain distribution differences in FDG, which may expose domain-specific feature patterns and enable attackers to infer sensitive information through side-channel analysis, increasing the risk of privacy leakage, particularly in scenarios with highly heterogeneous data [198]. To address these challenges, future research should explore dynamic, domain-aware privacy protection mechanisms that adapt to the unique characteristics and privacy needs of each domain [81]. It is also essential to investigate privacy-enhanced models that leverage domain alignment techniques to minimize interdomain discrepancies while safeguarding domain-specific information [44]. Finally, promising potential to design privacy-aware models that reduce reliance on direct data access while enhancing cross-domain generalization capabilities [199].

C. Heterogeneity

Heterogeneity is a core challenge in FDG, which can be divided into two main aspects: intradomain heterogeneity and interdomain heterogeneity [50]. Intradomain heterogeneity refers to the differences in data distribution or features within the same domain, such as imbalanced data distribution, variations in client capabilities, and privacy constraints, which lead to overfitting during local model training and impact the global generalization ability. Interdomain heterogeneity, on the other hand, refers to the significant differences in data distribution or features across different domains, such as inconsistent feature spaces, label distribution shifts, and domain interference, which increase the difficulty of cross-domain generalization. These two types of heterogeneity not only increase the complexity of model training but also raise the demands on the model's generalization ability. Furthermore, there is an interactive effect between intradomain and interdomain heterogeneity, which further exacerbates the complexity of FDG. Future research should focus on developing intradomain data augmentation and personalized learning methods to mitigate intradomain heterogeneity, designing domain-invariant feature learning and domain-weighted aggregation strategies to reduce interdomain differences [50], and exploring hierarchical

learning frameworks [200] and dynamic adjustment mechanisms [201] to collaboratively address both intradomain and interdomain heterogeneity issues. By addressing these challenges, FDG can better adapt to the complex data environments of the real world, providing stronger model generalization capabilities for cross-domain applications.

D. Communication-Efficient FDG

Communication efficiency in FDG becomes more complex due to clients representing different domains. Specifically, FDG faces unique challenges, including interdomain redundancy, where clients transmit redundant or domain-irrelevant information, increasing communication overhead; the need for cross-domain generalization, which demands learning domain-invariant features while minimizing communication costs; dynamic domain alignment, as client domains may evolve, necessitating efficient methods to reduce frequent updates; and resource-constrained environments, where limited computational and communication capabilities of devices like mobile or the IoT further strain efficiency. Future research should prioritize domain-specific communication protocols to reduce redundancy [202], cross-domain model compression techniques to maintain generalization with lower data volume [159], efficient FDAL methods (e.g., anchor-based or distributed adaptation) to synchronize models with minimal updates [203], meta-learning to enable domain-invariant feature learning with reduced communication [201], and adaptive scheduling strategies to dynamically balance communication frequency and data volume based on domain importance and client resources [204]. Addressing these challenges will help reduce communication costs while enhancing model generalization capabilities.

E. Computation-Efficient FDG

Computation-efficient FDG aims to enable FDG systems to operate effectively in resource-constrained environments while maintaining robust generalization across diverse domains. Efficient computation in FDG is a critical challenge, particularly due to the increased computational demands of cross-domain generalization tasks and the resource constraints of client devices such as mobile or IoT devices. FDG must address the added complexity of learning domain-invariant features across multiple domains while operating within limited computational capabilities. Key challenges include the high computational complexity of cross-domain learning, uneven distribution of domain data across clients, and the need for dynamic adaptation to evolving domains. Future research should focus on developing lightweight techniques for efficient deployment on resource-constrained devices [205], designing selective domain participation strategies to involve only relevant clients in specific training rounds, exploring federated domain distillation methods to collaboratively learn compact and generalizable models [169], and investigating hardware accelerators optimized for cross-domain learning

tasks. Additionally, dynamic computation resource allocation strategies [206] should be developed to balance computational load based on domain relevance and device capabilities. Computation-efficient FDG contributes to the advancement of edge computing, enhances model scalability, and facilitates the practical deployment of applications.

F. Scalable FDG

Scalable FDG aims to enable FDG systems to perform effectively in large-scale settings with numerous clients, diverse domains, and resource constraints. It is a critical challenge for FDG, as it involves not only the number of clients but also the diversity of domains they represent. Future research should focus on developing scalable domain aggregation algorithms capable of handling thousands of clients representing distinct domains, designing dynamic domain clustering techniques to group clients with similar domains and reduce the complexity of model aggregation [207], addressing domain-specific client dropout and network failures to ensure robustness in large-scale multidomain federated settings, and exploring transfer learning [24], [59] and meta-learning [54], [146] approaches to enhance generalization capabilities across an increasing number of domains. By addressing these challenges, FDG can achieve scalable and efficient learning in dynamic, heterogeneous environments, enabling its application in real-world scenarios.

G. Continuous FDG

It aims to enable models to generalize effectively across unseen domains in dynamic environments where new domains are continuously introduced. A key challenge of this paradigm is adapting to dynamic domain shifts, where models must learn from evolving domain distributions without access to past domain data due to privacy constraints. Moreover, scalability emerges as a critical issue as the number of domains increases over time. Future research should focus on developing lifelong DG techniques [208] to enable models to adapt to new domains while maintaining performance on seen domains, investigating incremental DA methods [209] to seamlessly integrate new domain knowledge into FL, and exploring the stability-plasticity tradeoff in FDG to ensure that models can adapt to new domains without catastrophic forgetting [210]. Additionally, dynamic model updating strategies should be explored to balance the need for continuous adaptation with the constraints of federated communication and computation. Continuous FDG addresses the challenges posed by dynamically evolving domains, enabling models to maintain high performance, robustness, and generalization capabilities over long-term deployment.

H. FDG to Novel Categories

FDG to novel categories highlights the potential of FL systems to extend their generalization capabilities to

previously unseen categories absent from the training datasets of participating domains. Existing FDG algorithms often assume that the label spaces—defined as the set of potential categories—across different domains are uniform [211]. However, a more realistic and challenging scenario arises when the source and target domains possess both overlapping and distinct label spaces, introducing an augmented category gap that further complicates the generalization task [212]. This challenge is compounded by the need to learn category-invariant features while maintaining sufficient discriminative power, alongside the entanglement of domain- and category-specific variations within federated environments. Additionally, the scarcity of labeled data for novel categories, combined with privacy constraints that limit access to raw data, intensifies these difficulties. Addressing these issues requires future research to prioritize methods for disentangling domain and category features, utilizing self-supervised and semi-supervised learning approaches [213], and extending few-shot and zero-shot learning frameworks to federated settings [147]. Moreover, standardized evaluation benchmarks reflecting dynamic label spaces and heterogeneous distributions are essential for advancing research and reliably assessing model performance and generalization capabilities.

VI. CONCLUSION

FDG is a significant research area within ML, focusing on generalization learning and privacy-preserving in distributed scenarios. This article offers a comprehensive analysis of the field of FDG, encompassing theoretical foundations, existing methodologies, available datasets, practical applications, evaluation strategies, evaluation metrics, and benchmarks. Additionally, through a meticulous analysis of these methods, several prospective research challenges are identified, which can pave the way for future investigations in the field. This survey aims to offer valuable insights to researchers and serve as a source of inspiration for future advancements in the field.

APPENDIX A DATASETS

In this section, we first provide a summary on existing commonly used datasets in FDG. Following this, we discuss the popular applications for this research area. Table 3 provides an overview of several commonly used datasets in FDG. These datasets have been extensively utilized in various FDG applications, including handwritten digit recognition, object recognition, transportation imaging, medical imaging, person re-identification, industrial edge intelligence, and human activity recognition. They serve as valuable resources for evaluating and benchmarking the performance of FDG algorithms and methodologies. Here, they provide a detailed description of several widely used datasets commonly employed in research studies.

Rotated MNIST, a variant of the original MNIST, incorporates rotation transformations to the original handwritten

Table 3 Summary of Widely Used Datasets in FDG Across Diverse Application Domains According to Various Categories by Domain: Handwritten Digit Recognition, Object Recognition, Transportation Imaging, Medical Imaging, Person Re-Identification, Industrial Edge Intelligence, and Human Activity Recognition. For Each Dataset, the Number of Domains, Classes, and Samples Is Provided, Along With a Brief Description and Relevant References

Area & Dataset	#Domain	#Class	#Sample	Description	Ref.
Handwritten digit recognition					
<i>Digit</i>	4	10	203,695	MNIST, MNIST-M, SVHN, USPS	[150]
<i>Digits-DG</i>	4	10	219,289	MNIST, MNIST-M, SVHN, SYN	[19]
<i>Rotated MNIST</i>	6	10	70,000	M0, M15, M30, M45, M60, M75	[16], [106], [115], [163]
<i>Digit-Five</i>	5	10	215,695	MNIST, MNIST-M, Synthetic Digits, SVHN, USPS	[108], [124]
Object recognition					
<i>Office-Home</i>	4	65	15,588	Art, Clipart, Product, Real-world	[19], [69], [99], [106], [115], [123], [124], [126], [163], [170]
<i>PACS</i>	4	7	9,991	Art, Cartoon, Photos, Sketches	[16], [19], [69], [99], [106], [115], [124], [126], [150], [163], [170]
<i>VLCS</i>	4	5	39,305	Ascal, LabelMe, Caltech, Sun	[16], [99], [126], [170]
<i>DomainNet</i>	6	345	586,575	clipart, infograph, painting, quickdraw, real, sketch	[106]
<i>Office-31</i>	3	31	4,652	Amazon, Webcam, DSLR	[123]
<i>Office-Caltech-Home</i>	7	65	17,964	Amazon, Webcam, Caltech, Art, Clipart, Product, Real-World	[16]
<i>Office-Caltech10</i>	4	10	2,533	Caltech, Amazon, Webcam, DSLR	[108], [124]
<i>ImageCLEF-DA</i>	3	12	1,800	ImageNet ILSVRC 2012, Caltech-256, Pascal VOC 2012	[123]
<i>Amazon Review</i>	4	2	7,671	Books, DVDs, Electronics, Kitchen & housewares	[108]
Transportation imaging					
<i>Cityscapes</i>	50	19	3,475	Real photos taken in the streets of 50 different cities	[68], [140]
<i>IDDA</i>	105	16		A synthetic dataset in the field of self-driving cars	[140]
<i>Mapillary Vistas</i>		66	25,000	The dataset covers diverse scenes from different domains and cities around the world	[68]
Medical imaging					
<i>Skin Lesion</i>	7	7		HAM10000, Dermofit, Derm7pt, MSK, PH2, SONIC, UDA	[57]
<i>ISIC 2018</i>	7	7	10,015	Melanocytic Nevus, Melanoma, Benign Keratosis, Basal Cell Carcinoma, Actinic Keratosis, Vascular Lesion, Dermatofibroma	[171]
<i>Fed-ISIC2019</i>	6	8	25,331	Melanoma, Melanocytic nevus, Basal cell carcinoma, Actinic keratosis, Benign keratosis, Dermatofibroma, Vascular lesion, Squamous cell carcinoma, None of the others	[164]
<i>Dermofit Image Library</i>	10	10	1,300	A comprehensive collection of dermatology images for study, analysis, and diagnosis of skin conditions	[171]
<i>YawDD</i>	2	3		A dataset of videos, recorded by an in-car camera	[132]
Person Re-identification					
<i>Person Re-ID dataset</i>	8	7,270	93,796	DukeMTMCReID, Market1501, CUHK03-NP, PRID, CUHK01, VIPeR, 3DPeS, iLIDt.	[138]
Industrial Edge Intelligence					
<i>CWRU</i>	4	10	900	A widely used dataset for fault diagnosis and condition monitoring of rotating machinery	[41], [60], [76], [147]
<i>PU</i>	/	10	52,497	A 6203 bearing dataset, obtained from Paderborn University, includes artificially induced and real damages.	[147]
Human Activity Recognition					
<i>UCI Smartphone</i>	30	6	10,299	Walking, Walking_upstairs, walking_downstairs, Sitting, Standing, Laying	[135]

digit images (M0, M15, M30, M45, M60, and M75), serving as a benchmark dataset for evaluating the robustness and generalization ability of ML algorithms and models in the context of image recognition and computer vision tasks.

PACS is a demanding dataset for DG, comprising seven distinct object categories from four diverse domains (Art

Painting, Cartoon, Photo, and Sketch), intricate models to effectively generalize across significant domain discrepancies and variations in artistic styles, rendering techniques, and visual representations.

Office-Home, encompassing four distinct domains (Artistic, Clipart, Product, and Real-World), has gained widespread usage in DA and transfer learning, serving

as a valuable resource for evaluating models' ability to generalize across diverse visual domains that encompass stylized and real-world images.

Digits-DG is a notable dataset specifically designed for DG in digit recognition tasks, incorporating four distinct digit datasets (MNIST, MNIST-M, SVHN, and SYN), introducing variations in font styles and backgrounds to simulate real-world scenarios and complex models to generalize across diverse writing styles, backgrounds, and digit representations.

VLCS is a comprehensive benchmark dataset designed to bridge the gap between vision and language understanding tasks by providing a rich collection of visual scenes accompanied by linguistic descriptions (Ascal, LabelMe, Caltech, and Sun), enabling the exploration and development of models that integrate visual perception and linguistic reasoning.

CWRU is a widely recognized benchmark in ML and fault diagnosis, comprising vibration signals from diverse rotating machinery under different operating conditions and fault scenarios, providing researchers with a valuable resource for developing and evaluating fault diagnosis and condition monitoring algorithms.

APPENDIX B EVALUATIONS

In general, FDG can be evaluated under various strategies and metrics, which are further discussed as follows.

A. Evaluation Strategies

When evaluating FDG methods, diverse strategies can be employed to comprehensively assess the model's capacity to generalize effectively to previously unseen domains. Here is a summary of evaluation strategies commonly used for FDG.

- 1) *Leave-One-domain-Out Cross validation (LODO)* [18], [99]: It is an evaluation strategy where one domain is held out during training, allowing the model's performance to be assessed on unseen domains and providing insights into its generalization capabilities across federated settings.
- 2) *Test-Domain Validation Set (TEVS)* [215]: It refers to a subset of data from a specific domain that is used for model evaluation during the validation phase.
- 3) *Training-Domain Validation Set (TRVS)* [215]: It refers to a subset of data from the same domain as the training data that are used for model validation during the training phase.

B. Evaluation Metrics

Evaluation metrics in FDG aim to assess the performance and generalization ability of models across different domains within the FL setting.

- 1) *Accuracy*: It is a prevalent evaluation metric employed to quantitatively assess the performance of ML models. Except for average accuracy, previous work [216]

followed [217] using the average accuracy among all classes and the average accuracy of the unknown classes in federated open-set DA. The average accuracy among all classes (OS)

$$\text{Acc (OS)} = \frac{1}{\mathbb{K} + 1} \sum_{k=1}^{\mathbb{K}+1} \frac{|x^t \in \mathcal{D}_k^T \wedge \hat{y}_k = y_k|}{|\mathcal{D}_k^T|}. \quad (34)$$

The accuracy of the unknown classes (OS*)

$$\text{Acc (OS}^*) = \frac{1}{\mathbb{K}} \sum_{k=1}^{\mathbb{K}} \frac{|x^t \in \mathcal{D}_k^T \wedge \hat{y}_k = y_k|}{|\mathcal{D}_k^T|}. \quad (35)$$

The average accuracy among known classes (UNK)

$$\text{Acc (UNK)} = \frac{|x^t \in \mathcal{D}_{\mathbb{K}+1}^T \wedge \hat{y}_{\mathbb{K}+1} = y_{\mathbb{K}+1}|}{|\mathcal{D}_{\mathbb{K}+1}^T|} \quad (36)$$

where $\hat{y}$ denotes the predicted value, $\mathcal{D}_k^T$ represents the set of target samples with the label y_k , and $\mathcal{K} := \{1, \dots, k, \dots, \mathbb{K}\}$ and the unknown class is defined as class $\mathbb{K}$. However, accuracy may not be suitable for imbalanced datasets (non-IID in FL) where the class distribution is skewed, as it can be influenced by the majority class. Therefore, it is important to consider other evaluation metrics.

- 2) *Mean Absolute Error (MAE)* [218]: In FDG, the MAE serves as a critical metric to evaluate the accuracy of models across distributed domains

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |y_k - \hat{y}_k| \quad (37)$$

where $\hat{y}$ denotes the predicted value, y denotes the actual observation, and N is the numbers of samples. In FDG, where models are trained on data from multiple domains without sharing raw data, MAE provides a valuable assessment of how well a model generalizes across different sites or domains.

- 3) *Standard Deviation (SD)* [126]: In FDG, the SD can be used to assess the distributional differences or variations across different domains.
- 4) *Dice Coefficient (Dice)* [18]: The Dice coefficient, also known as the Sørensen–Dice coefficient, is a similarity metric commonly used in image segmentation tasks and medical image analysis. It measures the agreement or overlap between two domains. The Dice coefficient is calculated as the ratio of twice the intersection of the two domains to the sum of the sizes of the individual domains

$$\text{Dice}(\mathcal{P}, \mathcal{Q}) = \frac{2 * \text{Comm}(\mathcal{P}, \mathcal{Q})}{|\mathcal{P}| + |\mathcal{Q}|} \quad (38)$$

where $\text{Comm}(\mathcal{P}, \mathcal{Q})$ represents the number of common samples between the two domains, while $|\cdot|$ represents the total number of samples in each domain. This metric is widely applied in tasks like image segmentation to assess the accuracy and similarity of segmented domains against a ground truth or reference. It offers a valuable metric for evaluating model performance in precisely delineating domains of interest, proving especially beneficial in scenarios characterized by class imbalance or an uneven distribution of domains.

- 5) *Hausdorff Distance (HD)* [18]: It is a metric used to measure the dissimilarity or distance between two domains of points, contours, or shapes. It quantifies the maximum distance between any point in one domain and its nearest point in the other domain. The HD is defined as follows:

$$\begin{aligned} \text{HD}(\mathcal{P}, \mathcal{Q}) &= \max(h(\mathcal{P}, \mathcal{Q}), h(\mathcal{Q}, \mathcal{P})) \\ h(\mathcal{P}, \mathcal{Q}) &= \max_{p \in \mathcal{P}} \left\{ \min_{q \in \mathcal{Q}} \|p - q\| \right\} \\ h(\mathcal{Q}, \mathcal{P}) &= \max_{q \in \mathcal{Q}} \left\{ \min_{p \in \mathcal{P}} \|q - p\| \right\} \end{aligned} \quad (39)$$

where $\mathcal{P}$ and $\mathcal{Q}$ are the two domains being compared and $h(\mathcal{P}, \mathcal{Q})$ represents the directed HD from domain $\mathcal{P}$ to domain $\mathcal{Q}$. It is calculated by finding the maximum distance from each point in $\mathcal{P}$ to its nearest point in $\mathcal{Q}$. Similarly, $h(\mathcal{Q}, \mathcal{P})$ represents the directed HD from domain $\mathcal{Q}$ to domain $\mathcal{P}$.

- 6) *WD* [159]: It is a mathematical metric that quantifies the dissimilarity between two probability distributions. In FDG, the WD measures the dissimilarity or discrepancy between the probability distributions of different domains in the FL setting. The WD is calculated as follows:

$$\text{WD} = \frac{1}{M^2} \sum_m \sum_{m'} \widehat{\text{WD}}(x_m, f_{m' \rightarrow m}(x_m)) \quad (40)$$

where M denotes the number of domains. Each $\widehat{\text{WD}}$ [219] is computed with the Sinkhorn algorithm.

- 7) *FID Score* [159]: It is a metric used to evaluate the quality of generative models, particularly in GANs. In FDG, it can be used as an evaluation metric to provide a quantitative measure of the dissimilarity between the generated samples from different domains. The FID is defined as follows:

$$\text{FID} = \frac{1}{M^2} \sum_m \sum_{m'} \widehat{\text{FID}}(x_m, f_{m' \rightarrow m}(x_m)). \quad (41)$$

The FID [220] could be utilized to assess GAN performance and effectively quantify disturbance levels.

- 8) *A-Distance* [164], [216]: *A*-distance (also known as the adversarial distance) is a metric used in DG and DA tasks, which measures the discrepancy between probability distributions of different domains or datasets

$$\hat{d}_A = 2(1 - 2\varepsilon) \quad (42)$$

where ε represents the generalization error of a two-sample classifier trained on the binary task of distinguishing input samples between the source and target domains.

- 9) *The SD of the mIoU* [140]: By computing the mean and SD of the mean IoU, the average performance and the variability in performance across multiple evaluations can be accessed. The mean provides an overall measure of accuracy, while the SD indicates the consistency or instability of the results.

- a) *Mean (mIoU)*: $\text{mIoU} = (1/N) * \sum_{i=1}^N (\text{IoU}_i)$, where N is the total number of evaluations and IoU_i represents the intersection over union value for the i th evaluation.
b) *SD*: $\text{SD} = ((\sum_{i=1}^N ((\text{IoU}_i - \text{mIoU})^2)/N))^{1/2}$.
c) *The SD of mIoU*: $\text{mIoU} \pm \text{SD}$.

- 10) *Group Effect (GE)* [108]: It evaluates negative transfer caused by inefficient model aggregation in FL, aiming to quantify the impact of client data differences on diverse local model updates that ultimately lead to negative transfer in the parameter space

$$\begin{aligned} \text{TTA}_f(\mathcal{G}_t) &= \frac{\sum_{(x,y) \in D_T} \mathbb{1}\{\arg \max_j \mathcal{F}(x; \mathcal{G}_t)_j = y\}}{|D_T|} \\ \text{GE}_t &= \frac{1}{M} \sum_{i \in 1,2,\dots,M} \text{TTA}_f(\mathcal{G}_t + \Delta_t^{(i)}) + \text{TTA}_f(\mathcal{G}_{t+1}) \end{aligned} \quad (43)$$

where $\mathcal{F}(x)$ is the neural network classifier, $\mathcal{F}(x)_j$ denotes the j th element of $\mathcal{F}(x)$, D_T represents the target domain dataset, $|D_T|$ denotes the size of the target domain dataset, $\mathcal{G}_t$ is the global model at time step t in the target task, and $\Delta_t^{(i)}$ is the update of domain i .

APPENDIX C BENCHMARKS

We present a comprehensive summary of existing benchmarks for evaluating the performance of FDG algorithms. Our analysis covers benchmarks in three categories: centralized DG, FL, and benchmarks specifically designed for FDG. Also, the detailed summary of these evaluations and benchmarks can be found in the following and in Table 4.

- 1) **DANN** [221] is a neural network architecture designed to accomplish precise classification of source data while simultaneously learning feature

Table 4 Summary of Evaluations and Benchmarks for DG, FL, and FDG. This Table Presents Representative Methods for Each Paradigm, Along With Their Adopted Evaluation Strategies and Performance Metrics

Paradigm	Method	Strategy	Metrics
DG	DANN [202]	TEVS	Accuracy
	JiGen [203]	LODO, TRVS	Accuracy
	Epi-FCR [204]	TEVS	Accuracy
	MTSSL [205]	LODO	Accuracy
	EISNet [206]	LODO, TRVS	Accuracy
	L2A-OT [8]	LODO	Accuracy
	DSON [207]	LODO	Accuracy
	Mixstyle [208]	LODO	Accuracy
FL	RSC [209]	LODO	Accuracy
	FedAvg [210]	TRVS	Accuracy
	FedProx [211]	TRVS	Accuracy
	Scaffold [212]	TRVS	Accuracy
FDG	Moon [47]	TRVS	Accuracy
	FedDG [18]	LODO	Dice, HD
	CSAC [16]	LODO	Accuracy
	FedADG [99]	LODO	Accuracy
	COPA [19]	LODO	Accuracy
	FedHealth [135]	TEVS	Accuracy
	FedIG(-A) [126]	LODO	Accuracy, SD
	FADH [44]	LODO	Accuracy
	CCST [69]	LODO	Accuracy
	FedINB [159]	TEVS	WD, FID
	FOSDA [197]	TEVS	OS, OS*, UNK
	FedDrive [140]	TEVS	mIoU $\pm$ SD
	FedKA [108]	LODO	GE
	FOSDA [197]	LODO	$\mathcal{A}$ -distance

representations that exhibit invariance across multiple source domains.

- 2) **JiGen** [222] is a supervised framework that leverages jigsaw puzzles as a training task to learn effective generalization across diverse visual domains.
- 3) **Epi-FCR** [223] is a scheme that learns domain shift using episodic training.
- 4) **MTSSL** [224] is a method that facilitates the learning of transferable features by employing a self-supervised task focused on predicting Gabor filter bank responses.
- 5) **EISNet** [225] is an innovative network that synergistically integrates self-supervised learning and metric learning approaches, effectively enhancing classifier performance, specifically in target domains.
- 6) **L2A-OT** [8] is a method that leverages synthetic data augmentation to learn domain-invariant features, facilitating effective generalization across different domains.
- 7) **DSON** [226] is a novel scheme that effectively enhances the generalization performance on target domains by integrating batch normalization and instance normalization methods.
- 8) **Mixstyle** [227] is an innovative method that combines features from different source domains to generate synthetic source domains, comprehensively enabling optimization of model generalization.
- 9) **RSC** [228] is an approach that selectively discards dominant features present in the training data to optimize the generalization capability of a model.
- 10) **FedAvg** [229] (federated averaging) is a distributed learning algorithm commonly used in FL settings without any generalization technique.
- 11) **FedProx** [230] is a novel framework that tackles heterogeneity in federated networks by serving as a generalized and reparametrized version of FedAvg.
- 12) **Scaffold** [231] is a novel algorithm that utilizes control variates to effectively address “client-drift” and minimize communication rounds, demonstrating resilience to data heterogeneity and client sampling while leveraging data similarity for faster convergence.
- 13) **Moon** [47] is an innovative FL framework, employing model-level contrastive learning to tackle the issue of heterogeneity in local data distribution.
- 14) **FedDG** [18] (FDG) is a solution that aims to learn a federated model such that it can directly generalize to completely unseen domains.
- 15) **CSAC** [16] is a novel privacy-preserving method for the separated DG task.
- 16) **FedADG** [99] (federated adversarial DG) is a scheme that utilizes federated adversarial learning to solve the DG problem in FL for IoT devices.
- 17) **COPA** [19] (collaborative optimization and aggregation) is a decentralized approach for DG and multisource unsupervised DA, utilizing a collaborative optimization and aggregation process to construct a generalized target model without sharing data across domains.
- 18) **FedHealth** [135] is a pioneering FTL framework for personalized wearable healthcare, leveraging FL and TL to overcome cloud-based personalization limitations.
- 19) **FedIG(-A)** [126] presents a novel approach that integrates client-agnostic learning with a combination of local training using mixed instance-global statistics and zero-shot adaptation through estimated statistics for inference.
- 20) **FADH** [44] (federated adversarial domain hallucination) is an innovative FDG approach that prioritizes domain hallucination to generate samples, optimizing the global model’s entropy while minimizing the cross entropy of the local model.
- 21) **CCST** [69] is a novel DG method for in FL, enabling style transfer across clients without data exchange to promote uniform source client distributions and mitigate model biases by aligning local models with the image styles of all clients.
- 22) **FedINB** [159] is an innovative federated domain translation method that generates pseudodata specific to each client, offering potential benefits for multiple downstream learning tasks.
- 23) **FOSDA** [216] (federated OSDA) is an advanced federated algorithm that incorporates an uncertainty-aware mechanism to generate a global model by prioritizing source clients with high uncertainty while preserving high consistency.

- 24) **FedDrive** [140] is an innovative benchmark framework that includes real-world challenges of statistical heterogeneity and DG.
- 25) **FedKA** [108] (federated knowledge alignment) is an innovative FDG method that utilizes feature distribution matching and a federated voting mechanism to enable the global model to learn domain-invariant client features and refine its performance with target domain pseudo-labels. ■

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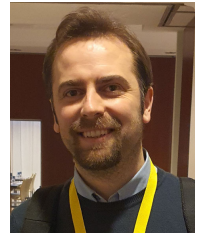
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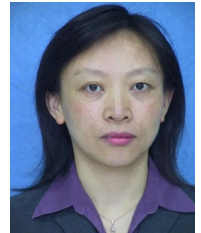
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